

Project Submission Pro-Forma

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- ☐ An abstract of the work completed
- ☐ A declaration of my contribution to the work and its suitability for the degree
- ☐ A table of contents
- ☐ A list of figures & tables (if applicable)
- ☐ A glossary of terms (where appropriate)
- ☐ A clear statement of my project objectives
- ☐ A full reference list (the Harvard referencing style is recommended for WMG)
- ☐ An appendix containing email confirmation of ethical approval or waiver

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**From Ingestion to Impact: Design and Evaluation
of an Evidence-First Multi-Agent AI Pipeline for
Auditable UK Early-Stage Portfolio Reporting and
Bounded Company Intelligence**

by

Faye Niu

Dissertation submitted in partial fulfilment for the Degree of Master of Science in
Applied Artificial Intelligence

WMG
University of Warwick

Submitted _____

Declaration

I have read and understood the rules on cheating, plagiarism and appropriate referencing as outlined in my handbook and I declare that the work contained in this assignment is my own, unless otherwise acknowledged.

No substantial part of the work submitted here has also been submitted by me in other assessments for this or previous degree courses, and I acknowledge that if this has been done an appropriate reduction in the mark I might otherwise have received will be made.

Project definition for my degree (as copied from the Specific Course Regulations for the course)

This substantial individual project is designed to strengthen your academic and research skills, while allowing you to explore a specific area of artificial intelligence that reflects your interests and career goals. The project will be presented as a written dissertation, submitted at the end of the academic year. You are expected to devote an appropriate portion of your time and intellectual effort to the project during the course i.e. approximately 600 hours over the course of the academic year.

My project relates to this definition in the following way

My project is an individual artificial intelligence project about creating public-company research with clear supporting evidence. I designed and built a local prototype for one user and prepared a study plan. The workflow starts with a Companies House number, which a named person must check. It saves public sources, exact passages, citations and time rules before it produces claims for review. OpenAI can help find possible public URLs, but the system then collects and checks each source through its own protected process. I have not claimed a completed live company-research run or a comparison with people. The workflow keeps extraction separate from verification, records missing, conflicting and unsupported evidence, and requires a named person to approve the current version before export. The dissertation reports the prototype, study plan, checked engineering evidence, limitations and remaining work. This project therefore supports my academic and research skills in an area of artificial intelligence that matches my interests and career goals.

Abstract

Early-stage portfolio reporting combines company submissions with public evidence that may be incomplete, delayed or inconsistent. The current process has three broad stages: company information is collected, aggregated into an executive summary, and prepared for review before use. The main risks are incorrect company identity, confusion between blank and zero values, evidence from the wrong period, unresolved conflicts and claims without adequate support. This dissertation evaluates an implemented evidence-first workflow that addresses these risks through explicit data rules, traceable source records and a separately recorded verification decision.

The prototype is a local Next.js dashboard over a private FastAPI and SQLite service. It preserves source class, reporting period, missing-data reason, file checksum and the exact link between each claim and its evidence. Planning, identity resolution, collection, extraction, normalisation, verification, composition and approval are separate operational roles with persisted hand-offs and bounded permissions. This role separation was selected to make production, verification and approval decisions inspectable. It was not selected on the assumption that several agents are generally better than one. A controlled public-web route was also implemented and tested with fake providers and local fixtures, but it was not run against a live company or website.

The executed evaluation used fourteen labelled fictional cases covering missing information, mixed data types and currencies, unclear identity, stale or inaccessible evidence, conflicts, repeated support, unsupported claims and prompt injection. Across three repeats, the role-separated condition with verification matched every expected decision. Precision, recall, F1 and source-record completeness were 1.000, and the unsupported-claim rate was 0.000. Without the separate verifier, precision was 0.455, F1 was 0.625, source-record completeness was 0.818 and the unsupported-claim rate was 0.545. The verifier accepted an observed zero with matching evidence, while holding a GBP 100 claim contradicted by GBP 200 evidence, a stale-period claim and an untrusted prompt-injection case. These results establish the programmed mechanism on designed cases.

The dissertation therefore demonstrates an inspectable evidence-first implementation and the observed effect of its verification gate in D0. It does not establish performance on real portfolios, superiority over a deterministic or single-agent design, reduced analyst effort, better investment decisions or production readiness. Manual reporting, participant review, live-web evaluation and managed-platform comparison were removed from the active research questions because the required observations were not available. They are retained as a prospective business pilot and future evaluation.

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Glossary and Acronyms

Term	Meaning
AI	Artificial intelligence.
API	Application programming interface.
C0–C3	Manual, deterministic baseline, multi-agent verification, and multi-agent plus human-review evaluation conditions.
D0–D2	Synthetic development, authorised pilot, and sealed final out-of-sample dataset tiers.
HITL	Human in the loop.
LLM	Large language model.
OOS	Out of sample.
RQ	Research question.
SSRF	Server-side request forgery.
URL	Uniform resource locator.

Introduction

1.1 Early-stage portfolio reporting and company intelligence

The verified project charter describes three broad reporting activities. Company information is collected, aggregated and analysed into an executive summary, then prepared for review before use. The portfolio/reporting operator receives company submissions and eligible public records. A company data contributor may supply internal values, a reviewer or approver decides whether the current report version may be released, and a downstream report user consumes the approved output. These are operational roles rather than assumed job titles. The legal company and reporting date must be clear because incorporation does not prove that a business is trading, and recent records may be limited by filing delays (Galanakis & Savagar, 2026; Hardman & Ramírez Santos, 2023).

Table 1.1. Verified portfolio-reporting process and main control point

Actor	Input	Activity and hand-off	Output	Pain point addressed
Data contributor / reporting operator	Company submission and eligible public records	Collect information and hand it to aggregation	Source records and typed observations	Mixed formats, uncertain identity, blanks and source gaps
Reporting operator	Typed observations and evidence	Aggregate and analyse, then hand a draft to review	Executive-level draft with claim links	Period mismatch, unsupported claims and hidden conflicts
Reviewer / approver	Draft, evidence, conflicts and gaps	Approve, reject or return the current version	Approved report for the downstream user	Unclear release authority and unrecorded corrections

Source note: Actor and process summary from `docs/PROJECT_CHARTER.md`. The repository evidence does not establish reporting frequency, labour cost, error rate or job title.

Evidence about a company’s development comes from dated events rather than one complete profile. Crowdfunding results vary by pitch, platform, country, sector, region and time. Published samples may also include only successful pitches (Estrin et al., 2024). Research on repeat funding uses the campaign round as its unit and reports links with investor numbers, targets, equity offered and overfunding on one platform (Wasti et al., 2024). UK funding data can connect opportunities, applications, decisions, organisations, projects and outcomes, but its coverage and links remain incomplete (Thorne et al., 2026). A report should keep each event tied to its date and source. It should not turn these observations into a score that claims to measure company quality.

These limits show that reliable reporting starts with the data, not with well-written text. A company register may be suitable for one task but not another. Legal identity and address fields are more available than management, financial or contact information (Krasikov et al., 2020). Even required Companies House fields can be empty, so quality rules must be clear, linked to the intended use and checked in code (Nikiforova et al., 2020). Extracting financial tables can also introduce errors through optical character recognition (OCR), repeated values, cell positions and missing negative signs (Bradley et al., 2024). The system must record the source, date, location and check result for each important statement. It must also show when evidence is missing or unclear.

1.2 Problem definition and motivation

The main data problem is to match different records to the correct legal company and to use only information that was available at the relevant date. Research that joined several registers used normalised names, alternative country names, approximate matching and manual review, but some records still could not be matched (Surak & Inkley, 2026). Companies House data can contain empty required fields and inconsistent country names (Nikiforova et al., 2020). Newer companies may also have no accounts because the filing date has not arrived (Hardman & Ramírez Santos, 2023). The prototype therefore waits until the company number and legal identity are clear. It keeps different types of missing data instead of changing blanks to zeros, and records when information first became available.

Text can sound convincing and still be wrong, even when it includes citations. Research on cited answers separates factual correctness from citation precision and recall. A citation marker alone does not guarantee any of these qualities (Gao et al., 2023a). A statement may also match a source that is itself wrong (Gao et al., 2023b). Public web pages can contain hidden instructions that try to change a model’s summary, source choice or tool actions (Greshake et al., 2023). For this reason, web search is used only to find possible sources. Each report claim must point to a saved source and an exact page or passage, and a separate check must confirm the support. Retrieved text is always treated as untrusted data.

Human approval creates a release boundary, but it does not guarantee a better result. General guidance for human-AI systems recommends explaining what the system can do, showing uncertainty and allowing people to correct or dismiss its output (Amershi et al., 2019). In one controlled task, some participants followed incorrect AI advice. Prompts that made them stop and think reduced this problem, but they also affected preference, trust and different participant groups in different ways (Bućinca et al., 2021). The prototype shows conflicts, missing evidence, limits and exact source locations, and it requires a named reviewer before export. The evaluation tests the evidence and verification mechanisms, while human outcomes remain a future pilot question.

1.3 Research aim

The aim of this dissertation is to design, implement and evaluate an evidence-first, role-separated workflow for early-stage portfolio reporting. The study examines how the prototype preserves evidence from intake to a reviewable report and how a separate verification stage changes claim admission in controlled cases. This follows design-science research, which links a practical problem to the development and evaluation of an artefact (Hevner et al., 2004; Peffers et al., 2007).

1.4 Research questions

Primary research question

How can an evidence-first, role-separated workflow produce traceable early-stage portfolio reporting claims, and what effect does a separate verification stage have on claim admission in controlled cases? The question concerns the implemented workflow rather than one model, agent label or interface. It is answered by combining RQ1 and RQ2 with the executed engineering and D0 evidence, as required in design-science evaluation (Hevner et al., 2004; Peffers et al., 2007).

RQ1: Input-to-claim accuracy and reliability

How does the implemented workflow preserve company identity, reporting time, missing states, source provenance and approval boundaries from input to report? RQ1 is answered with implementation records, engineering tests and labelled D0 cases. The assessment covers exact identity holds, value and period semantics, source-record completeness, claim-to-evidence links and repeated outcomes. These measures keep data quality tied to the reporting task and keep factual correctness separate from citation support (Gao et al., 2023a; Nikiforova et al., 2020).

RQ2: Effect of separate verification

What effect does a separately recorded verification stage have on supported-claim precision, unsupported claims, conflict handling, source-record completeness and repeatability in D0? RQ2 uses a paired comparison between C1, which has no separate verifier, and C2, which adds verification. Both conditions receive the same fictional cases and proposed values. This avoids treating a citation link or fluent statement as proof that a claim is supported (Gao et al., 2023a;b).

Scope narrowing

The original contract also proposed manual C0, named-review C3, live-web and managed-platform comparisons. They require authorised participants, frozen references, live access and recorded time, cost or review observations. Those observations were not available, so these comparisons are not active research questions. Human control may support correction and accountability, but it may also create delay or over-reliance (Amershi et al., 2019; Bućinca et al., 2021). Chapter 7 records this limitation once, and Chapters 6 and 8 specify the future pilot needed to test it.

Research objectives and evaluation plan

Four objectives connect the questions to the study. O1 defines shared rules for legal identity, dates, missing data and source records. O2 implements those rules in a workflow that stops when required evidence is missing or invalid. O3 evaluates the implemented controls and the C1/C2 verifier difference using the fixed D0 fixture. O4 distinguishes supported, unsupported, conflicting, unavailable and missing evidence in every result. Together, these objectives connect the artefact to observable evaluation criteria without treating successful execution as proof of practical benefit (Hevner et al., 2004; Peffers et al., 2007).

Table 1.2 summarises this evaluation plan and the evidence currently available.

Table 1.2. Research questions and evaluation plan

Question	How it will be answered	Main measures	Current evidence and limit
Primary question	Combine RQ1 implementation evidence with the paired C1/C2 D0 comparison.	Traceability; decision correctness; supported-claim precision; unsupported-claim rate; source-record completeness; repeatability.	Engineering checks and D0 answer the bounded question. No real-portfolio or business-benefit inference is made.
RQ1	Inspect the implemented records and test fixed identity, date, missing-data, provenance and approval rules.	Required records and states; exact claim links; source completeness; repeated outcomes.	Repository implementation, tests and D0 provide the evidence.
RQ2	Compare C1 without a separate verifier with C2 on the same cases and with the same information access.	Precision; recall; F1; unsupported-claim rate; contradiction handling; source-record completeness; repeatability.	Executed on fourteen fictional cases across three repeats.

Limit: Manual, participant, live-web and managed-platform comparisons are prospective work. Investment outcomes, production readiness and complete web coverage are not tested.

Source note: Developed by the author from the fixed evaluation plan and current evidence status. It does not add a study result.

1.5 Scope, contribution, and exclusions

The system is a local, single-user research prototype. It is not a deployed service and can be reached only from the same computer. Its purpose, user and operating limits are stated because risk management depends on the setting in which a system is used (National Institute of Standards and Technology, 2023). Public company information may be sent to an external model only after the required checks. Restricted or internal material stays local and is not sent to public generative-AI applications or APIs (Central Digital and Data Office, 2023). These limits do not prove legal compliance, production security or zero retention by a provider.

The contribution has three parts: an evidence-first prototype, an executed controlled evaluation and a reporting method that traces claims to sources. It combines design with tests against stated objectives, without treating implementation as proof of operational usefulness (Hevner et al., 2004; Peffers et al., 2007). It also keeps factual correctness separate from citation support, so each sourced claim still needs a page or passage check (Gao et al., 2023a;b). The public-web research route provides additional implementation and adversarial-test evidence, but it is not used to answer the active questions.

The project excludes buy or sell recommendations, speculative valuations, profiles of people, automatic publication and claims that the system is ready for remote, multi-user or production use. Financial communications can affect investment decisions and may create further duties. This supports a cautious scope, not a claim of legal compliance (Financial Conduct Authority, 2024). Public web content remains untrusted because hidden prompt injection can change model output or tool actions (Greshake et al., 2023). A named person must approve a report before export, which keeps human responsibility clear (Central Digital and Data Office, 2023). These controls define the prototype’s authority boundary; their wider limits are addressed in Chapter 7.

1.6 Dissertation roadmap

Chapter 2 moves from evidence risks to design criteria and reasonable architecture alternatives. Chapter 3 defines the executed evaluation and places unrun protocols in the appendices. Chapter 4 explains the implemented architecture and operational roles. Chapter 5 reports engineering and D0 results, including individual claim decisions. Chapter 6 interprets RQ1 and RQ2 and specifies a prospective business pilot. Chapter 7 is the authoritative location for ethics, governance and study limitations. Chapter 8 states the bounded contribution and future evidence. This order follows design science by linking the problem, artefact, evaluation and communication (Hevner et al., 2004; Peffers et al., 2007).

Literature Review

2.1 Scope and structure of the literature review

The literature relevant to this dissertation spans several connected fields rather than one established research stream. Studies of early-stage finance examine how funding rounds, public support and investor activity may signal company development, but their findings depend on the platform, sample and reporting context (Estrin et al., 2024; Wasti et al., 2024). Research on company registers also shows that public records can support analysis while remaining incomplete, delayed or difficult to link across sources (Hardman & Ramírez Santos, 2023; Thorne et al., 2026). These findings make legal identity, reporting date and source context central to reliable portfolio reporting.

A second body of work considers how raw information becomes a defensible report. Data-quality research treats accuracy, completeness, consistency and timeliness as qualities that depend on the intended use (Nikiforova et al., 2020). Work on dataset documentation and reproducible research stresses the need to explain origins, transformations, assumptions and limitations (Geburu et al., 2021; Pineau et al., 2021). Research on citation-supported generation further shows that factual correctness, citation support and citation completeness are different questions (Gao et al., 2023a). Together, these studies suggest that well-written output is not enough unless each important claim can be traced and checked.

The final body of literature concerns AI-assisted retrieval, multi-agent workflows and human control. Retrieval may improve access to relevant material, but it can also return incomplete, irrelevant or unsupported information (Gao et al., 2023c; Huang et al., 2023). Dividing work between agents can make roles and handovers explicit, although the use of several agents does not itself prove better results (Guo et al., 2024; Wu et al., 2024). Human-AI research similarly supports clear limits, uncertainty and opportunities for correction while warning that people may still rely on incorrect advice (Amershi et al., 2019; Buçinca et al., 2021). The sections that follow bring these themes together, moving from early-stage company evidence and data quality to traceable reporting, AI-assisted processing, human review and the research gap addressed by this dissertation.

2.2 Early-stage finance, portfolio reporting, and public company intelligence

Founders usually know more about an early-stage business than outside investors, and funding platforms affect what information is shown. Crowdfunding evidence from several countries shows that the

amount raised and investor participation vary by pitch, sector, region, country and platform. The sample included successful pitches rather than every attempt (Estrin et al., 2024). A study of Crowdcube also found links between later funding rounds, investor response and funding outcomes within one platform sample (Wasti et al., 2024). These observations can add context to a portfolio report, but investor numbers and completed rounds do not prove company quality or future performance.

Portfolio reports should record funding and public support as dated events with clear sources. In the Crowdcube study, an earlier successful round may act as a signal, but its meaning still depends on the campaign and platform (Wasti et al., 2024). UK research on public support found later differences between participating firms, but it also noted selection effects and did not claim that the support caused those differences (Mondal & Mellor, 2025). Rebuilding the UK funding process requires links between opportunities, applications, decisions, projects, organisations and outcomes, even though these records often lack shared identifiers (Thorne et al., 2026). A round, grant or award can support a dated fact. It cannot prove future success or provide a complete funding history.

These events need to be linked to the correct legal company. A Companies House number identifies a registered company, and incorporation data give timely and broad information about company creation. However, incorporation does not mean that a company is trading. The register includes dormant firms and excludes some unincorporated activity (Galanakis & Savagar, 2026). Analysis of the full register also depends on the snapshot date, company type and filing rules. Recent records are affected by filing delays (Hardman & Ramírez Santos, 2023). Open access alone does not make register data complete or suitable for every use (Krasikov et al., 2020). Register status and filings describe a legal record, not a complete view of business activity or finances.

This review defines permissible reporting claims; it does not create an investment model. Each accepted statement should identify the event, legal company, source, date, locator and supported content, while missing information, uncertain links and conflicts remain visible. Crowdfunding findings depend on sample and platform (Estrin et al., 2024), and public-support differences may reflect selection rather than causal effect (Mondal & Mellor, 2025). Financial communications should present relevant benefits and risks clearly and fairly (Financial Conduct Authority, 2024). The system therefore excludes buy or sell advice, speculative valuations, causal performance claims and a general company-quality score.

2.2.1 Sector importance and existing reporting approaches

The reporting problem sits within a material but uneven early-stage investment market. The British Business Bank records £10.8 billion of announced UK smaller-business equity investment across 2,048 deals in 2024, while deal numbers fell by 15.1 per cent and seed- and venture-stage activity weakened (British Business Bank, 2025). Monitoring also consumes professional attention: a survey of 885 institutional venture capitalists reported an average of 18 hours each week working with portfolio companies and about 60 per cent interacting with them at least weekly (Gompers et al., 2016). These descriptive results come from different periods and settings; monitoring is broader than reporting, and neither source measures the effect of a reporting tool.

Private-market evidence also limits what reporting can consolidate. Kaplan and Lerner describe commercial venture-capital datasets as incomplete and inconsistent, with differences in definitions, coverage of failed firms, backfilling and non-reporting (Kaplan & Lerner, 2016). The British Business Bank similarly treats angel-investment evidence as partial because much activity is private and reporting is patchy (British Business Bank, 2025). Portfolio managers, analysts and downstream report users therefore need dates, definitions, legal identities and visible coverage gaps. These sources establish a

traceability problem; they do not establish the time, error, cost or satisfaction benefit of automation.

Several technical approaches are plausible. Manual documents and spreadsheets preserve flexible judgement, but their provenance and consistency depend on the local practice. Deterministic data pipelines can apply repeatable identity, time and quality rules, although lineage must be explicit and schema changes can invalidate those rules (Buneman et al., 2001; Nikiforova et al., 2020). Retrieval-augmented generation can extend unstructured search but may retrieve irrelevant or hostile material, while one fixed process or several bounded roles can apply the same final support rule (Gao et al., 2023c; Guo et al., 2024). The reviewed sources do not measure this project’s manual workflow, so effort and error cannot be assigned to any approach before a matched study.

Commercial products cover related but non-equivalent workflows. Visible describes portfolio-data collection, monitoring, dashboards and investor reporting; Zapflow describes integrated deal and portfolio records, monitoring and reporting; and Affinity describes relationship-centred activity capture, dashboards and reports (Affinity, n.d.; Visible, n.d.; Zapflow, n.d.). These are current vendor descriptions, not independent tests. Their public pages do not establish claim-level support accuracy, early-stage UK suitability, reviewer effort, time savings or superiority. Table 2.4 therefore compares documented functions and unresolved evidence without scoring products or treating this prototype as proven better.

Table 2.1. What early-stage company evidence can and cannot support

Evidence class	Fact shown by the source	Careful report statement	Claim not supported	Key local literature
Crowdfunding / funding round	Attributed platform or publisher record of a campaign or round, with its date, disclosed stage or amount, and source scope where stated.	“The named source records this entity’s campaign or funding event on the stated date.” Include the exact locator and qualify platform or sample scope.	Proof of firm quality, universal investor preference, causal impact, future performance or investment merit.	estrin2024digital wasti2024 successive
Public grant or state-support record	Named programme, recipient, award or support event, date, publisher and explicit amount/currency where available.	“The publisher records the named award or support event for this legal entity.” State the date, locator, cut-off and any amount-coverage limit.	That support caused innovation or growth; that the recipient is superior; complete support history or measured impact.	mondal2025state thorne2026 funding fca2024 promotions
Companies House identity / filing event	Exact company number plus a legal name, status, incorporation or filing event from a dated register snapshot.	“At the stated cutoff, the register records this legal fact or filing for company number X.” Retain filing period and register scope.	Incorporation as active trading; complete operations, ownership or financial health; private funding, valuation or current state beyond the cutoff.	galanakis2026chrt hardman2023 small krasikov2020 ready
Reconstructed funding-lifecycle link	Conservatively linked events whose component sources, identifiers, dates, linkage rule and unmatched gaps remain visible.	“These attributed records form a partial reconstructed timeline for the same legal entity under the stated linkage rule.”	Exhaustive funding history; absence as proof of no event; a causal pathway, universal lifecycle or forecast of success.	thorne2026 funding mondal2025state

Meaning: Developed by the author. This is not a result, score, ranking, valuation, recommendation or cause-and-effect model.

Source note: The local sources are estrin2024digital, wasti2024successive, mondal2025state, thorne2026funding, galanakis2026chrt, hardman2023small, krasikov2020ready and fca2024promotions. Supporting pages are in claim-ledger rows 2.2-P1–2.2-P4.

2.3 Data quality, company identity, missing data and time

Data quality depends on the task. It is not one fixed property of a dataset. Accuracy, reliability, timeliness, completeness and consistency need rules that apply to the relevant field or record (Nikiforova et al., 2020). Research on company registers supports this view. Open and accessible data may still be incomplete, difficult to interpret or unsuitable for a business task (Krasikov et al., 2020). A portfolio report should therefore explain how each field will be used and which quality checks matter. One combined quality score could hide a serious problem in a particular field and cannot prove that the data suit every purpose.

Company matching also depends on the source. An exact company number from the official register is the main legal identifier; a company name is only a possible match. Funding sources without shared identifiers need careful exact or approximate matching, and errors and gaps still remain (Thorne et al., 2026). Work across several registers also required name normalisation, approximate comparison and manual checks when company or country names disagreed (Surak & Inkley, 2026). Companies House identifies incorporated companies rather than all economic activity (Galanakis & Savagar, 2026). An uncertain match should therefore stay unresolved until a named reviewer checks it. Records with similar names must not be merged automatically.

Missing data must keep its meaning instead of becoming a default number. Documentation should state what is missing, why it is missing and what is still unknown (Gebu et al., 2021). A financial table contains values, empty cells, positions and different types of information. An empty cell does not automatically mean zero (Bradley et al., 2024). Completeness and validity are also separate qualities that need their own checks (Nikiforova et al., 2020). This project distinguishes a blank, zero, none, not applicable, not reported, not found publicly, invalid and observed value. Not found publicly does not mean that something does not exist. These categories are a project design choice, not a general standard from the literature.

The system also needs separate dates. It records when an event happened, when the information became available, when it was collected and the last date allowed in the report. Register snapshots and filing delays show why recent facts may not have been available at an earlier reporting date (Hardman & Ramírez Santos, 2023). Incorporation and later business activity also happen at different times (Galanakis & Savagar, 2026). Using information published after the reporting date creates data leakage and may make an evaluation look better than it should (Kapoor & Narayanan, 2023). A correction needs a new version and date. It must not be used to support an earlier report.

Together, these rules create a clear and testable evidence model. It records company identity, the type of missing value, event date, availability date, source, page or passage and version. Invalid, conflicting and unresolved records remain visible. The checks must match the user's task and the data being checked (Nikiforova et al., 2020). Documentation should show unknowns, errors, dependencies and corrections (Gebu et al., 2021). The evaluation must also prevent data leakage and unclear comparisons (Kapoor & Narayanan, 2023). These sources explain the method used later in the dissertation. They do not prove that the prototype was completed or report its results.

Table 2.2. Data meanings that must remain separate

Data control	Required distinction	Unsafe combination	Effect on reporting or evaluation	Key local literature
Legal identity	Keep an exact source-scoped legal identifier separate from names, domains and aliases; hold unresolved candidates or conflicts.	Name-only or fuzzy auto-merge; treating incorporation as economic activity.	Evidence can attach to the wrong company, contaminating reports and identity accuracy or coverage measures.	thorne2026funding surak2026gateways galanakis2026chrt
Observed value / missing state	Separate a value from the reason it is missing. The project distinguishes observed, zero, blank, explicit none, not applicable, not reported, not found publicly and invalid.	Treating null, blank, no record, invalid and zero as equivalent.	False totals or absence claims follow; blank-zero and missing-state evaluation becomes invalid.	gebru2021datasheets bradley2024synfintabs nikiforova2020quality
Publication / effective / retrieval / cutoff time	Retain value period, publication or effective availability, retrieval time and reporting cutoff separately.	One timestamp; retrieval treated as publication; later evidence used at an earlier cutoff.	Leakage across time, old or future claims and an unfair historical evaluation. Evidence from the wrong date must be held.	hardman2023small galanakis2026chrt kapoor2023leakage
Correction / version / snapshot	Keep write-once source bytes and checksum, data meaning, schema and source versions, and each later correction as a new dated version.	Overwrite the earlier snapshot or silently backfill a correction into a past report.	Claims and calculation totals cannot be repeated. Historical comparisons and links to approval become invalid.	gebru2021datasheets hardman2023small kapoor2023leakage
Use-case data quality	Assess explicit dimensions and executable rules against the named data object and reporting task.	One context-free quality score, or assuming open availability means fitness.	The cause of failure becomes unclear, eligibility rules are applied wrongly and results across tasks cannot be compared fairly.	nikiforova2020quality krasikov2020ready

Meaning: Developed by the author and not a result. The exact list of missing states is a project choice, not a published or universal standard.

Source note: The local sources are thorne2026funding, surak2026gateways, galanakis2026chrt, gebru2021datasheets, bradley2024synfintabs, nikiforova2020quality, hardman2023small, kapoor2023leakage and krasikov2020ready. Supporting pages are in claim-ledger rows 2.3-P1–2.3-P5.

2.4 Source records, source checks, conflicts and traceable reporting

Provenance records where a statement came from and how it was processed. Foundational database work distinguishes why a query result exists from where contributing values were located (Buneman et al., 2001); this explains lineage, not free-form report truth. Documentation should record purpose, contents, processing, gaps, errors, dependencies and corrections (Gebru et al., 2021), while repeatability requires sufficient data, code, steps and measures (Pineau et al., 2021). Each portfolio statement should therefore link to its saved source, exact locator, transformations, checks and approval, keeping traceability distinct from source quality, truth and completeness.

Checking a source and checking a claim are two different steps. The source check covers access, source type, permission to use it, date and whether a stable copy can be saved. The claim check asks whether an exact passage supports the statement. Citation support and completeness are different from factual correctness, and a visible reference may be irrelevant or only partly useful (Gao et al., 2023a). Retrieved material may be ignored or contradicted, and it may be quoted correctly even when the source is wrong (Gao et al., 2023b). These decisions and files must be recorded so that the work can be repeated (Pineau et al., 2021). A search snippet, bare URL or fluent summary is not enough to pass either check.

When permission and technology allow it, an accepted source should be saved as a fixed local copy. Its record should include a checksum, collection time, exact page or passage and version. A correction creates a new version instead of replacing the old one. Dataset guidance supports recording outside resources, distribution, updates, errors and older versions (Gebru et al., 2021). Repeatable research also needs enough saved files and procedures (Pineau et al., 2021). Model cards similarly separate the version, intended use, evaluation method and limits (Mitchell et al., 2019). These fields are project choices, and some sources cannot legally or technically be saved.

Conflicting evidence must be retained. Hallucination research distinguishes factual conflict, unverifiable claims and overstatement, linking reliable output to uncertainty and checking against trusted knowledge (Huang et al., 2023). A source link does not resolve disagreement when text conflicts with evidence or the source is wrong (Gao et al., 2023b); nor can one citation marker represent correctness, passage support and completeness (Gao et al., 2023a). Supported disagreements remain grouped and unverifiable claims unresolved until a stated rule or named reviewer decides them. The system must not automatically prefer the majority, newest or most fluent version.

A traceable report records source discovery, checking, capture and use, together with the claim, support decision, conflict, approval and export. Measurement should state uncertainty, comparison, tested conditions and independent-review limits (National Institute of Standards and Technology, 2023). A separate verifier checks evidence without rewriting the claim because correctness, support, completeness and attribution differ (Gao et al., 2023a;b). Human-facing systems should expose uncertainty and permit correction or dismissal (Amershi et al., 2019). Only supported, approved claims may be exported. These principles motivate the design; they do not demonstrate verifier or reviewer benefit, user utility or deployment safety.

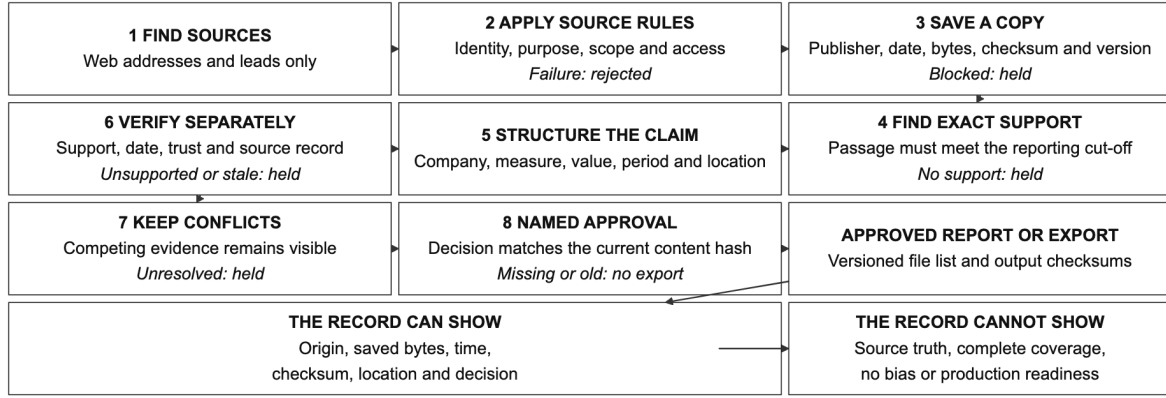


Figure 2.1. How a possible source becomes an approved report claim. Developed by the author from the cited literature and fixed project rules. The figure does not show measured results.

2.5 LLM-assisted source finding and evidence extraction

Retrieval-augmented generation (RAG) separates document indexing, retrieval and text generation, but it does not guarantee an answer supported by evidence. Retrieved text also does not prove source quality, completeness or truth. Retrieval may miss relevant material, select irrelevant passages or lead to unsupported answers (Gao et al., 2023c). It can reduce gaps in a model’s knowledge, but incorrect material can still produce misleading output (Huang et al., 2023). A correct link to a source is also different from a correct answer (Gao et al., 2023b). A large language model (LLM) can help find possible sources, but finding, checking, extracting and verifying them must remain separate steps.

The saved Responses API documentation describes model-managed `web_search`, inline citations, annotations, domain filters and a list of consulted sources (OpenAI, n.d.-b). These features can support search planning and a list of possible sources (Gao et al., 2023c). However, the settings do not guarantee a number of sources or citations, and visible citations may still be incomplete or unsupported (Gao et al., 2023a). The documentation shows what the API offered when the page was saved. It does not show how this prototype performed. Returned URLs, snippets, annotations and model text remain possible sources until they pass the source checks.

Structured evidence should first be extracted with fixed, rule-based code. Unstructured public or synthetic evidence may use an optional model, but only to suggest a claim that links to an exact passage. Factual correctness, passage support and citation completeness remain different measures (Gao et al., 2023a). A correct source link also does not prove a claim when the evidence disagrees (Gao et al., 2023b). Both routes end with fixed checks of the exact passage, data structure and format, followed by separate verification. This reflects the need to test relevance, support, rejection and reliability separately (Gao et al., 2023c). Unsupported content is rejected or left unresolved. It is never filled in from model memory.

Public content found through search must be treated as untrusted data, not as instructions. Indirect prompt injection can hide harmful instructions in searched material and change summaries, queries, source choices or later actions (Greshake et al., 2023). RAG can also receive noisy, conflicting or hostile text (Gao et al., 2023c). The prototype therefore limits tool permissions, checks each destination, separates extraction from other actions, limits redirects and requests, and stops safely when a check fails. Testing, uncertainty, separate review and safe failure are recognised risk controls (National Institute of Standards and Technology, 2023). The literature supports using these controls, but it does not prove that they are effective in this prototype.

The report should record how much search was allowed rather than claim that the whole web was searched. Search context, token limits, live access, domain filters and source lists can be configured. Search may also be optional, and the number of sources is not guaranteed (OpenAI, n.d.-b). Evaluation should keep factual correctness, citation support and completeness separate (Gao et al., 2023a). It should also test relevance, support, rejection, how parts work together and reliability (Gao et al., 2023c). A repeatable method records queries, sources, reporting date, token and cost limits, then lists areas that were inaccessible, missing, conflicting or not verified (Pineau et al., 2021). It cannot measure how much of the whole web was found.

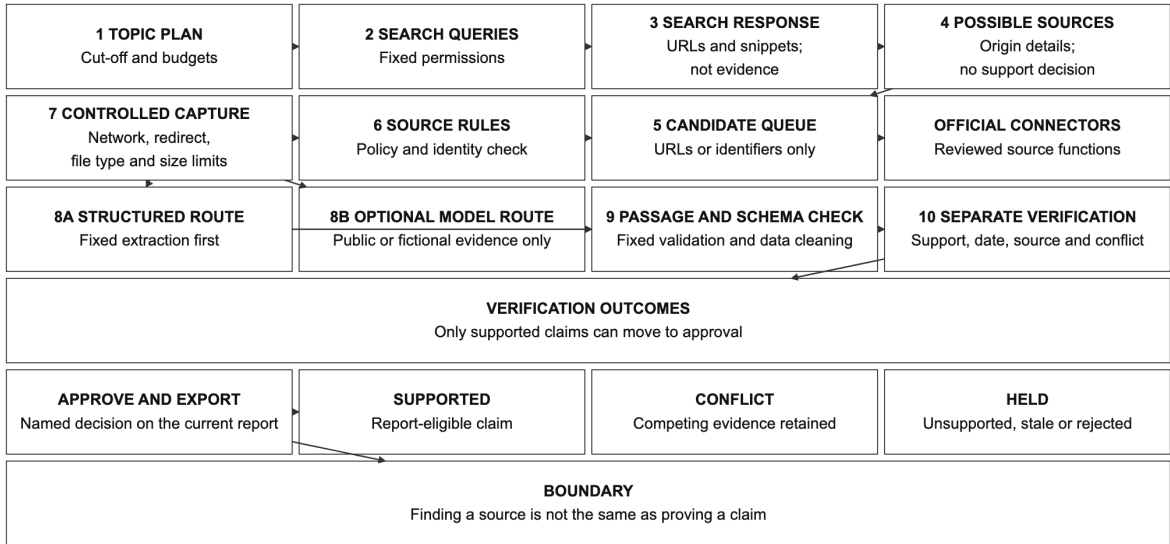


Figure 2.2. Why finding a source is not the same as having evidence. Developed by the author from the cited literature and fixed project rules. The figure does not show measured results.

2.6 Functional role separation and human approval

A multi-agent workflow divides a task between programmed roles. This does not prove that the roles have collective intelligence. Systems in the literature use different interfaces, role descriptions, communication structures and messages (Guo et al., 2024). AutoGen also shows that developers choose the roles, tools, human input, control flow and stopping rules (Wu et al., 2024). Table 2.3 therefore gives each role one clear responsibility, defined inputs and outputs and a visible failure limit. A role name or conversation does not prove ability, independence or benefit. Design-science evaluation must test whether the system meets its objectives (Peffer et al., 2007).

The workflow therefore needs ordered stages, limits and stop rules. Incorrect content can pass between roles, while extra roles add messages, work and cost (Guo et al., 2024). Analysis of 1,642 traces from seven multi-agent systems identified fourteen failure modes spanning system design, inter-agent alignment and task verification, with rates varying across selected systems and tasks (Cemri et al., 2025). Much of the wider dataset used model-assisted annotation, and its coding, mathematics and general-agent tasks differ from portfolio reporting. Programmed control flow, tool permissions, human-input conditions and stops remain necessary (National Institute of Standards and Technology, 2023; Wu et al., 2024). Handovers, retries and stop reasons should be visible; a separate verifier may hold a claim without rewriting it. More roles are not presumed better.

Human review gives a person authority to decide; it does not create new evidence. The interface should show the claim, exact support, conflicts, uncertainty, limits and content hash. A named reviewer can then approve, reject or return it for correction. Human-AI guidance supports starting, dismissing and correcting AI output and providing overall controls (Amershi et al., 2019), but this is not proof of a better outcome. Explanations may not stop people from relying too much on AI. One controlled task found trade-offs between performance, usability and effects on participant groups (Bućinca et al., 2021). Human oversight must match the use and level of risk (National Institute of Standards and Technology, 2023). Accuracy, trust and usefulness cannot be claimed before an approved study.

Evaluation must retain failures and limits. Multi-agent systems can propagate incorrect content and face coordination, scaling and benchmark limitations (Guo et al., 2024); AutoGen’s flexible roles and pilots do not establish universal superiority (Wu et al., 2024). Logs should record role, task, attempt, input, output, hash, time, tokens, errors, stopping, cancellation and recovery. Fixed cases should cover each role and the full workflow, including disagreement and safe failure, with methods and measures reported for repetition (National Institute of Standards and Technology, 2023; Pineau et al., 2021). Agent, message or conversation counts are not task-success measures.

Research on no-code agent tools makes a managed implementation a credible future alternative. AutoGen Studio exposes models, tools, memory and workflows through a visual interface and includes evaluation and debugging functions. Its authors identify architecture comparison and reliability as open research problems, and state that the tool is not production-ready. Current SpaceXAI documentation describes Grok Bot as persistent named Bots with distinct jobs, tools and sources, working rules, approval boundaries and visible hand-offs. These sources justify evaluating Grok Bot as a possible substitute for part of the custom orchestration layer. They do not show that it is easier to maintain, more accurate or suitable for portfolio reporting (Dibia et al., 2024; Guo et al., 2024; SpaceXAI, 2026).

Table 2.3. What each workflow role may and may not do

Role	Permitted input / tools	Required output	Stop / failure condition	Forbidden authority
Query planner / discovery	Fixed question, cut-off, source rules and limits; approved search or catalogue interfaces.	Structured search plan and possible-source queue with origin, scope and stop reason. Possible sources are suggestions only.	Scope, time, attempt or tool limit reached; no eligible candidate; policy mismatch. Record empty, missing or held state.	Accept a source; treat a suggestion as evidence; create a report claim; widen permissions or approve actions.
Evidence extractor	Accepted write-once evidence, exact location and structured schema; fixed parser or approved extractor with clear limits.	Structured extraction proposal with exact passage, value or state, date and source record.	No exact support; invalid schema; ambiguity, conflict, stale evidence, budget or tool failure. Abstain or hold.	Fill gaps from memory; change source rules; hide missing information; decide support, report eligibility or approval.
Independent verifier	Producer record, write-once evidence or passage, source record and fixed support rules.	Separate assessment recording supported, contradicted, stale, insufficient or rejected-untrusted, with reason.	Evidence or location absent; value, period, trust or source mismatch; unresolved conflict or rule failure. Reject or hold.	Silently rewrite the producer record; seek replacement support; override policy; compose, approve or export.
Fixed report writer	Verified eligible claims, clear exceptions, agreed template, ordering and citation rules.	Repeatable draft report with citations, conflicts, unknowns, limits and content hash.	Any claim lacks an eligible verification outcome; template, schema or hash invariant fails. Refuse composition.	Add or infer facts; repair missing evidence; alter verification; hide exceptions; approve or publish.
Named human reviewer	Versioned report, exact support, verifier records, conflicts, uncertainty, limitations and content hash.	Named, timed approve, reject or return-for-correction decision with rationale and report version.	Identity or version mismatch; unresolved support or conflict; incomplete evidence; no review authority. Do not approve.	Manufacture support; silently change the approved version; override source eligibility; treat review as proof of truth or utility.

Meaning: Developed by the author and not a result. Programmed roles and messages do not show independent intelligence, better performance or benefit.

Authority rule: Model-assisted roles may suggest or assess records within set limits. The verifier cannot quietly rewrite a claim, the report writer cannot add facts, and the reviewer adds authority to decide rather than new evidence.

Source note: The local sources are guo2024multiagent, wu2024autogen, peffers2007dsrm, nist2023airmf, amer-shi2019guidelines, bucinca2021forcing and pineau2021reproducibility. Supporting pages are in claim-ledger rows 2.6-P1–2.6-P4.

2.7 Main findings from the literature and research gap

The literature does not establish one best reporting architecture. Private-market datasets have coverage and definition limits, while lineage and citation research separate origin, support and truth (Buneman et al., 2001; Gao et al., 2023a; Kaplan & Lerner, 2016). Vendor pages describe different monitoring, relationship and reporting products but provide no independent comparison (Affinity, n.d.; Visible, n.d.; Zapflow, n.d.). The bounded gap is evaluative: whether identity, missingness, time, provenance, verification and human-authority controls work together under shared early-stage inputs. It is not a novelty claim. Table 2.4 makes the alternatives and unresolved evidence explicit.

For RQ1, the established problem is that company and funding records can be incomplete, delayed and difficult to link, while an exact citation can still support an incorrect or partial statement (Gao et al., 2023b; Kaplan & Lerner, 2016; Thorne et al., 2026). The unresolved issue is end-to-end transformation: whether fixed identity, time, missing-value and source-support rules preserve the intended meaning and abstain on unsupported cases. RQ1 therefore requires labelled transformation outcomes, claim support, provenance completeness and repeat reliability rather than a demonstration that a report can be generated (Nikiforova et al., 2020; Pineau et al., 2021).

For RQ2, role decomposition can make responsibilities visible, but surveys and trace analysis also identify coordination, verification and system-design failures; a single process with the same fixed gate remains a credible alternative (Cemri et al., 2025; Guo et al., 2024). The unresolved issue is whether a separately recorded verifier changes admission decisions under equal inputs, and at what cost. RQ2 therefore compares conditions with and without that verifier using supported-claim precision, recall, unsupported-claim rate, abstention, failures, time, tokens and cost (Peffer et al., 2007; Pineau et al., 2021). A favourable result cannot be attributed to agent labels without a comparison that isolates the rule and separation.

For future human evaluation, guidance supports correction and decision authority, while controlled evidence also shows usability and over-reliance trade-offs (Amershi et al., 2019; Bućinca et al., 2021). Commercial pages describe dashboards and reporting workflows, but not independently measured reviewer effort or decision quality (Visible, n.d.; Zapflow, n.d.). Whether named review changes correctness, completeness, editing effort or elapsed time therefore remains a pilot question. An implemented approval screen cannot answer it.

Public-web research remains a separate engineering case. The saved web-search documentation describes controls for finding possible sources but does not show how this prototype performed (OpenAI, n.d.-b). Retrieved content also creates risks from prompt injection and weak source support (Gao et al., 2023a; Greshake et al., 2023). Figure 2.2 therefore separates finding a source from accepting it as evidence. The live comparison between official sources alone and LLM-assisted search was not approved or fixed before this evaluation. It is not an extra research question or a result, which follows the need to report only what the evidence supports (Pineau et al., 2021). Chapter 3 turns these gaps into the fixed evaluation method.

Table 2.4. Architecture alternatives for the reporting problem

Approach	What it can provide	Main trade-off for this prototype	Evidence position
Manual work-flow	Flexible human collection, reconciliation and judgement	Provenance, timing and review depend on the local procedure	No measured baseline in this study
Deterministic pipeline	Repeatable rules, typed data and inspectable lineage	Limited handling of new unstructured evidence; wrong rules repeat consistently	Used for structured processing, not evaluated as a complete alternative
Single grounded agent	One interaction path can retrieve, extract and apply a fixed support gate	Production and checking share context and permissions unless separated explicitly	Credible alternative; not compared empirically
Role-separated workflow	Narrow permissions, persisted hand-offs, failure isolation and an approval boundary	More states, messages, storage and maintenance	Selected for inspectability; D0 isolates the verification gate, not general multi-agent superiority
Managed agent platform	Configured persistent roles, tools, context and hand-offs without building every interaction surface	Provider-managed boundaries and shared computer state require separate control review; setup and maintenance effects remain unknown	Grok Bot is a future comparator; not implemented or evaluated

Source note: Author synthesis from Buneman et al. (2001), Nikiforova et al. (2020), Gao et al. (2023c), Guo et al. (2024), Wu et al. (2024), Cemri et al. (2025), Dibia et al. (2024) and SpaceXAI (2026). The table compares design choices; it does not report an effectiveness comparison.

Research Design and Methodology

3.1 Applied design-science strategy

This dissertation uses an applied design-science strategy because the research object is a working portfolio-reporting artefact for a bounded organisational problem. Design science links construction to evaluation against stated objectives (Hevner et al., 2004; Peffers et al., 2007). The method therefore separates three kinds of evidence: implementation records show what was built, engineering tests show which specified behaviours were exercised, and D0 shows how two workflow conditions handled labelled fictional cases. None of these observations is treated as evidence of business benefit.

The student built and evaluated the artefact, which creates confirmation risk. The main controls were fixed case labels and metrics, versioned code and fixtures, saved result files, complete failure accounting and repeated runs. Reproducibility requires enough code, data, procedure and reporting detail for another person to inspect the route to a result (National Institute of Standards and Technology, 2023; Pineau et al., 2021). These controls make changes visible, but they do not provide independent validation.

Figure 3.1 shows the evidence boundary used throughout the study. Manual work, participant review, live-web operation and managed-platform comparison were removed from the active research questions because they had no authorised observations. Their protocols are retained only as future evaluation material. This keeps unobserved outcomes null rather than converting an evaluation plan into a result (Hevner et al., 2004; National Institute of Standards and Technology, 2023).

3.2 Evaluand and comparison conditions

The evaluand is the complete local workflow configuration: input rules, catalogue, source access, operational roles, verification, prompts, schemas, computing environment and stop conditions. This binds each observation to one defined artefact version. Design-science evaluation requires comparison with stated objectives (Hevner et al., 2004; Peffers et al., 2007), while system reporting should state the intended use, version and evaluation procedure (Mitchell et al., 2019).

The executed comparison uses C1 and C2. C1 applies the shared catalogue and normalisation rules and emits each eligible cleaned value unless identity is unclear. It retains available provenance but has no separately recorded verifier. C2 receives the same candidate values and evidence, then admits a claim only when the separate fixed verifier marks it as supported. Composition cannot override that decision. Each condition ran three times on the same fourteen D0 cases, so the measured difference is

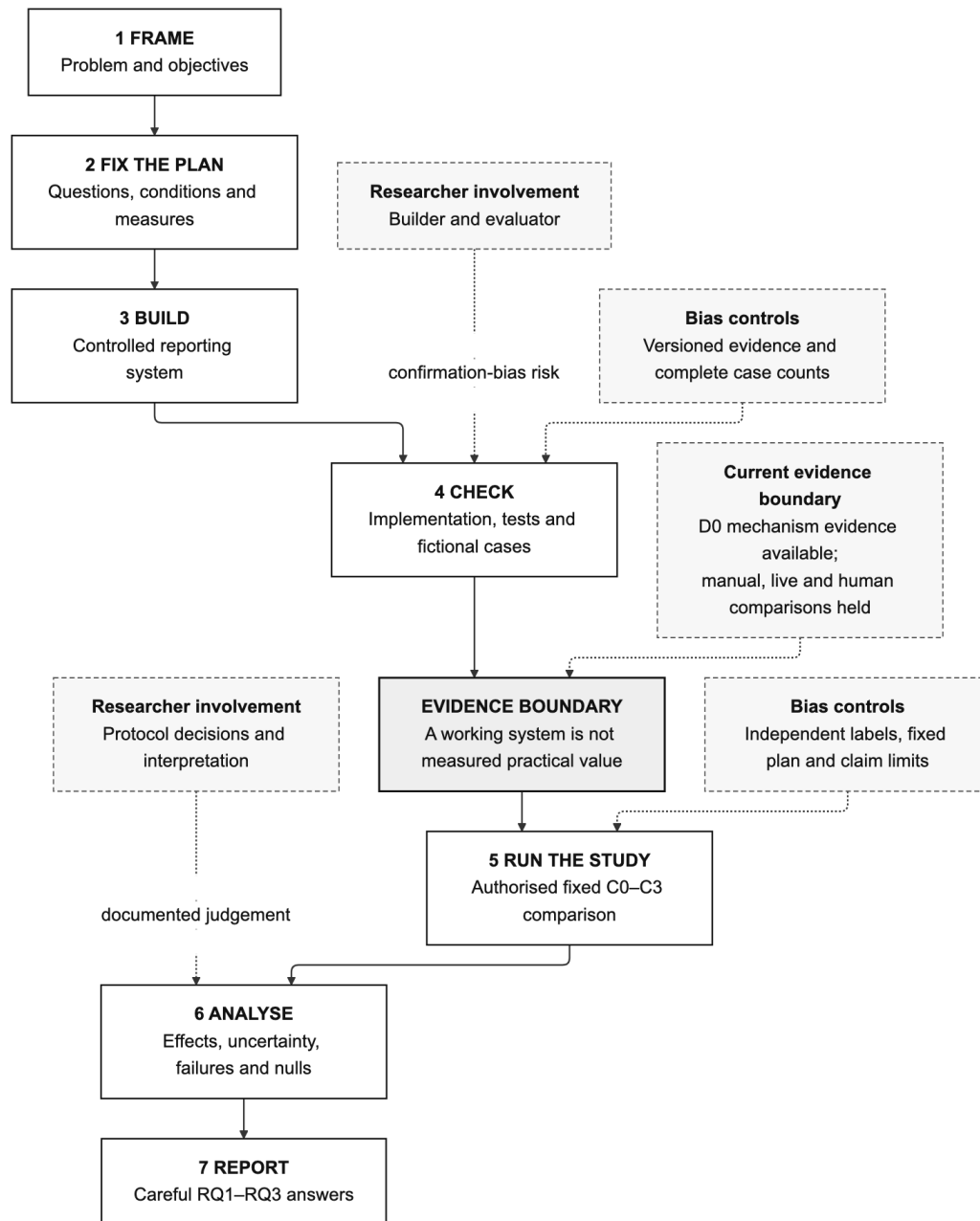


Figure 3.1. Boundary between construction, engineering checks and study results. Developed by the author from the cited methods literature and project rules; it is not a result.

the verification gate and its recorded decision, not a claim about agent personalities (Guo et al., 2024; Wu et al., 2024).

The original protocol also defined C0 manual work and C3 named review. These conditions did not run and are not used to answer RQ1 or RQ2. Their detailed condition matrix is kept in Appendix D to support a future authorised pilot. Retaining the protocol preserves the planned comparison without presenting missing observations as evidence (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

3.3 D0 fixture, labels and freeze

D0 contains fourteen visible, labelled fictional cases designed to expose known error categories. They cover complete and sparse records, blank and observed-zero values, mixed types, conflicting and stale evidence, unsupported and inaccessible sources, unclear identity, duplicate support, narrative evidence, mixed currencies and prompt injection. Synthetic task material is useful when it provides controlled labels, but its construction and permitted use must be explicit (Bradley et al., 2024; Gebru et al., 2021).

The fixture and its evaluation record are checksum-pinned. Loading checks the schema version, data classification, source version, file checksum and test-only namespace. Case and evidence identifiers are placed in a benchmark namespace that the operational collector rejects. The labels specify the expected emission decision and verification status. The complete case ledger is in Appendix F (Gebru et al., 2021; Pineau et al., 2021).

D0 is a development fixture, not a sample of companies. The rules and cases were developed together, so the result establishes only whether the programmed gate behaves as designed on these cases. A future pilot must create independent labels before a sealed final set is opened. Separating development from final evaluation reduces leakage risk (Kapoor & Narayanan, 2023), and versioned data records support later repetition (Mitchell et al., 2019; Pineau et al., 2021).

3.4 Source parity and reference decisions

For D0, both conditions receive the same candidate value, evidence record, reporting period and precondition. Scoring uses the fixture’s expected emission and status fields. The evaluator does not give either condition later evidence or a wider source window. This equal-input design is sufficient for the narrow C1/C2 mechanism comparison, but the visible development labels are not an independent reference (Kapoor & Narayanan, 2023; Pineau et al., 2021).

A real-company evaluation would require an eligible-source list, cut-off date, independently labelled claims and recorded adjudication. Agreement would indicate consistency rather than truth because reviewers can share an error (Artstein & Poesio, 2008). Exact locators and source records would also remain necessary because citation presence does not establish complete or correct support (Gao et al., 2023a; Pineau et al., 2021). The full prospective reference-parity matrix is moved to Appendix D.

3.5 Measures and denominator rules

The primary D0 measures are supported-claim precision, recall, F1, unsupported-claim rate, supported-claim rate, source-record completeness, verification or abstention accuracy, contradiction accuracy and exact repeat consistency. Precision uses supported emitted claims over all emitted claims. Recall uses correctly emitted required claims over all required claims. Unsupported claim rate uses unsupported emitted claims over all emitted claims. These denominators keep claim correctness separate from citation presence and support (Gao et al., 2023a;b).

Schema and normalisation accuracy are also reported because both conditions share earlier processing. Contradiction accuracy has two applicable cases and is interpreted as a small case result, not a stable rate. Duration values below one millisecond were truncated to zero, so they cannot support a speed comparison. Model cost is zero because D0 uses fixed local rules. Measures without D0 labels remain null, not zero. Clear missing states and denominators are necessary for task-specific data-quality interpretation (Geburu et al., 2021; Nikiforova et al., 2020).

The complete metric and denominator matrix in Appendix F also defines prospective human time, edit, usability, source-coverage and cost measures. Those fields are pilot metrics, not results of this study (Gale et al., 2013; National Institute of Standards and Technology, 2023).

3.6 Analysis and interpretation

The evaluator reports counts before rates and keeps all fourteen paired cases in both conditions. The first repeat supplies the scored values, and the complete ordered result list is compared across all three repeats. Individual cases explain why an aggregate changed. No confidence interval or hypothesis test is reported because D0 is a small, deliberately constructed fixture rather than a sample drawn from a target population (Demšar, 2006; DiCiccio & Efron, 1996).

Error analysis uses the expected verification states and records false emission, false withholding, conflict, stale evidence, inaccessible evidence, identity hold and untrusted content separately. Negative and unavailable outcomes remain in the record. Any later study should pair the same company-periods, report exclusions before rates and use uncertainty at the company or report level, not treat repeated fields as independent observations (Kapoor & Narayanan, 2023; Pineau et al., 2021).

3.7 Ethics, leakage prevention and reproducibility

Every input is classified as restricted, internal, public or synthetic. Restricted material remains local, and public availability does not itself authorise every use. Participant observation and live external processing require separate authority, purpose, consent and storage decisions (Central Digital and Data Office, 2023; Geburu et al., 2021; National Institute of Standards and Technology, 2023). D0 uses only fictional data and no external source or model call.

The evaluation record retains the Git revision and dirty-state warning, platform and dependency versions, migration version, fixture and protocol checksums, test commands, result files, failures and output hashes. A checksum detects byte changes but cannot establish truth or independent replication (Mitchell et al., 2019; Pineau et al., 2021). The detailed implementation snapshot is in Appendix G.

Collection or evaluation must stop when authority is unclear, a credential is exposed, unexpected

personal data appears, source terms would be breached, restricted data would reach an unapproved service, or a sealed evaluation set is compromised. The detailed evidence-control matrix is retained in Appendix E. These controls define the research boundary; they do not imply ethics approval for the future pilot (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

System Design and Implementation

4.1 Requirements and trust boundaries

The implemented release is a local, loopback-only, single-user research prototype, not a production service. Requirement NFR-SEC-001 limits the server binding, while FR-UI-003 rejects non-local clients and unexpected host headers. For the bounded company-research route, FR-INT-001 allows a Companies House number without a name or website; however, FR-RES-001 requires that identifier to receive named review before research starts. The number is therefore the sole required intake value for this route, not proof of complete company identity or economic activity (Galanakis & Savagar, 2026; Hardman & Ramírez Santos, 2023; National Institute of Standards and Technology, 2023).

Data is classified as public, internal, restricted or synthetic at intake, and each case retains its purpose and actor. Under the charter, the student operates the prototype, portfolio companies retain confidentiality interests, and data owners or the university ethics body authorise restricted or participant use. FR-EXT-005 makes external-model processing opt-in and limits it to public or synthetic evidence. The implemented client sends `store=False`, but provider documentation still describes abuse-monitoring and application-state retention unless approved controls apply; this setting is not Zero Data Retention (Central Digital and Data Office, 2023; Gebru et al., 2021; OpenAI, n.d.-a).

Outputs require review. FR-REP-005 blocks report export without approval, and FR-RES-007 requires named review plus content-hash checking for company-deck access. These tests establish mechanisms, not reviewer benefit. The scope excludes remote or multi-user production deployment, person profiling, speculative valuation and automatic buy/sell recommendations. Direct live Companies House retrieval remains held at G2; the public-web route is implemented, but neither its live smoke nor future frozen evaluation has run. Table C.1 maps these controls and holds against misleading communication and untrusted content, without proving compliance or production safety (Financial Conduct Authority, 2024; Greshake et al., 2023; National Institute of Standards and Technology, 2023).

The full requirements and trust-boundary matrix is provided in Appendix C.

4.2 Architecture and local deployment

The implemented artefact is a two-process modular local application. Next.js renders the sole operator dashboard; its server-side proxy calls a private FastAPI JSON API and approved-deck download surface, so the browser does not contact FastAPI directly. API handlers call a `Runtime` that assembles importing, source, workflow, reporting, intake and optional company-research services. The CLI con-

structs the same runtime for database initialisation, imports, workflow execution, review and export. This separates presentation from domain actions and persistence, but implementation alone does not demonstrate utility (Hevner et al., 2004; Peffers et al., 2007).

SQLite stores canonical identity, workflow, claim, review and audit records with foreign keys enabled. Startup runs Alembic to the declared migration head rather than creating tables directly. Imported bytes and source captures are written once below ignored local directories, checksum-verified and recorded by path; approved exports use staged, manifest-backed bundles. Separating relational meta-data from local evidence bytes supports traceability, while recorded versions, dependencies and environments aid reproducibility rather than proving the evidence true (Geburu et al., 2021; Pineau et al., 2021).

Native development runs the Next.js dashboard and FastAPI service as separate local processes. Docker Compose mirrors that topology in pinned application and Python images: only Next.js is published on host loopback, while its server-side proxy reaches FastAPI on the private Compose network. Both roots are read-only; the API alone mounts the persistent `portfolio-state` volume. Health, proxy, Host/client, CSRF and migration checks exercise these mechanisms. Figure 4.1 maps the request and data path. The public-web route is implemented behind explicit gates, but its live smoke and evaluation remain unrun (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

This deployment remains a single-user research prototype. It has no production authentication or authorisation, tenant isolation, horizontal scaling, managed secret or storage service, durable job system, backup regime, central observability or demonstrated disaster recovery. Loopback, CSRF, local files and container restrictions reduce the current local attack surface; they do not establish hosted security or resilience. Any remote or multi-user release would require a separately designed, authorised and evaluated operating model (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).



Figure 4.1. Local architecture, deployment and external-boundary status. Author synthesis of repository contracts, implementation and local verification evidence; conceptual and non-empirical.

4.3 Shared data definitions, storage and source records

Canonicalisation starts from a versioned catalogue. Each metric definition fixes a key, type, unit, sourceability, aliases and one period semantic; the catalogue hash prevents a version from being reused with altered meaning. Reporting periods and observations then bind company, metric, period and raw submission. Normalisation preserves original and normalised values separately. Blank, explicit zero, none stated, not applicable, not reported, not found publicly, source unavailable and invalid are typed states, not interchangeable nulls. This improves task-specific interpretability but embeds project definitions that still require domain approval (Geburu et al., 2021; Nikiforova et al., 2020).

SQLite separates canonical companies and identifiers from submissions, observations, reporting periods, evidence, claims and reports. Foreign keys, uniqueness constraints and indexes protect relationships, while Alembic revisions 0001–0010 reproduce the ORM schema and incrementally add provenance, company-research and reviewed group-scope records. Tests compare migrated and metadata schemas and exercise downgrade preflights. This approach supports a local prototype and reversible evolution, but many semantic states remain strings or JSON checked by application services; database consistency therefore cannot establish source correctness, business meaning or concurrent production use (Geburu et al., 2021; Pineau et al., 2021).

Imported files are hashed before persistence and published create-once under checksum-addressed paths. Connector snapshots record source and version, reviewed identifier, cutoff, locator, media type, checksum, retrieval and publication times, and classification. Facts retain value, unit, currency, periods, effective and publication times, exact structured locator and extraction method/schema. A versioned derivation hash covers terminal snapshot metadata, facts and events, detecting same-byte parser drift. The public-web case implements redacted capture and checksum verification, but no live capture has run. Immutability and hashes strengthen repeatability and tamper detection; they neither prove truth nor guarantee source access (Geburu et al., 2021; Pineau et al., 2021).

Claims persist normalised values and link many-to-many with evidence items; each evidence item carries publisher, locator, times, checksum, connector version, classification and trust or time flags. A separate verification stage records supported, contradicted, insufficient, stale or untrusted outcomes without rewriting the source. Portfolio reports can export these links without raw evidence. The separate company-research schema stores accepted verbatim spans with source URLs and groups divergent multi-source statements as contradiction candidates requiring named resolution. That path is live-unrun, and a citation or checksum provides traceability rather than correctness; human resolution may also remain fallible (Gao et al., 2023a;b).

Report sections are append-versioned. An edit retires the current section, increments report and optimistic-lock versions, clears approval and recalculates the report-content hash. Approval records named actor, reason and report version only after every claim has verification; export then requires the approved current version, creates JSON, Markdown and HTML with a content hash and file manifest, stages writes and re-verifies hashes before reuse. This binds the decision to controlled content without autonomous publication. Gold labels, reviewer outcomes and other evaluation fields remain protocol-only and null pending authority; implementation does not demonstrate review benefit. Figure 4.2 maps these boundaries (Amershi et al., 2019; National Institute of Standards and Technology, 2023; Pineau et al., 2021).

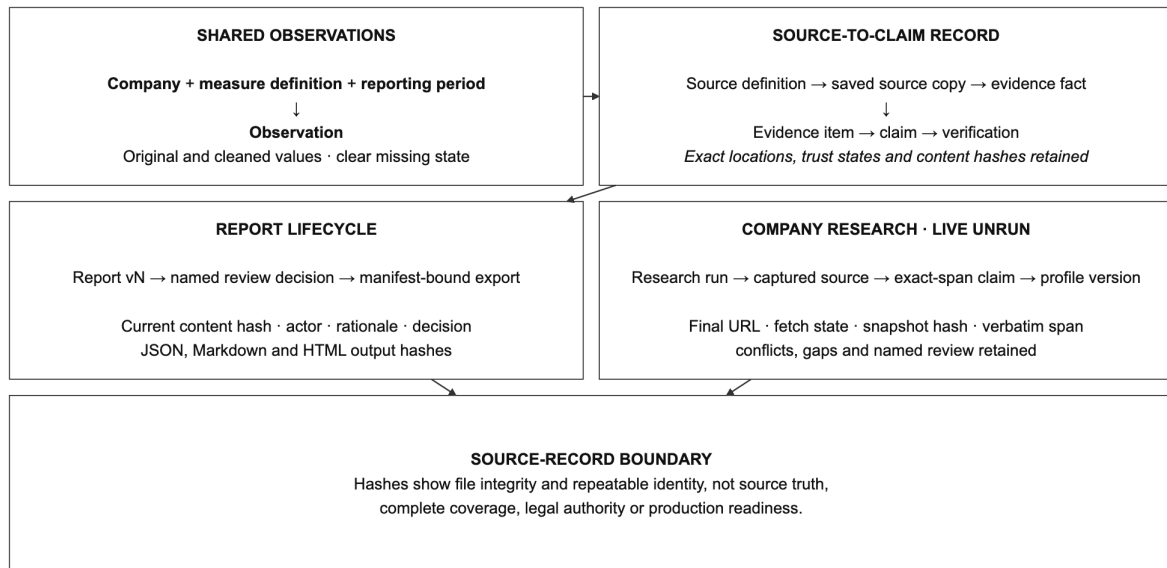


Figure 4.2. Canonical data, provenance and approval lineage. Author synthesis of implemented relational contracts and held protocol extensions; conceptual and non-empirical.

4.4 Intake and legal-identity control

The portfolio-import route accepts XLSX, CSV and JSON, then hashes the original bytes before writing a create-once snapshot. Its dataset identifier combines those bytes with the reporting-period label; a repeated hash and period reuses the stored submission only when classification, cutoff and profile remain compatible. Each company row then enters the same deterministic identity resolver, while conflicting identifiers become explicit counted holds rather than silent repairs or merges. This keeps source bytes, period and identity decisions traceable, but a checksum supports repeatability and integrity, not factual truth (Gebru et al., 2021; Pineau et al., 2021).

The company-intelligence route accepts a Companies House number without a name or website. It removes spaces and hyphens, uppercases prefixes and zero-pads short numeric values before structural validation. For a newly submitted number, valid shape alone creates an exact identifier, unresolved placeholder company and identity-held case; it does not query or confirm against the register. Website, name-plus-jurisdiction and declared-company document routes instead persist submitted claims. Domains remain pending, and document bytes remain untrusted and checksum-bound, because legal registration and operational identity are not equivalent (Galanakis & Savagar, 2026; Hardman & Ramírez Santos, 2023).

Identity becomes ready only after a named actor accepts the exact identifier with a substantive reason before source use. A rejection leaves identity unresolved; an identifier/name disagreement, a reused name with another number, a cross-classification collision, or a domain verified elsewhere is refused or held. Names may suggest a candidate, but neither name normalisation nor fuzzy similarity merges companies; name-only portfolio imports create unresolved records for review. This conservative choice sacrifices recall to reduce false joins, consistent with evidence that heterogeneous registers need cautious linkage and retain matching error (Galanakis & Savagar, 2026; Surak & Inkley, 2026; Thorne et al., 2026).

Normalised intake content, purpose, classification and contract versions form a fingerprint, so exact repeats reuse one artefact and case; database uniqueness backs that local rule. Document creation tolerates an identical concurrent file winner and rejects checksum disagreement. However, the pre-query and insert path is a synchronous local boundary, not demonstrated multi-worker exactly-once processing. Downstream source collection requires the reviewed number to match the source key and company. Figure 4.3 maps these gates. Direct live Companies House access remains held at G2; offline replay does not establish current registry truth (Galanakis & Savagar, 2026; Gebru et al., 2021; Pineau et al., 2021).

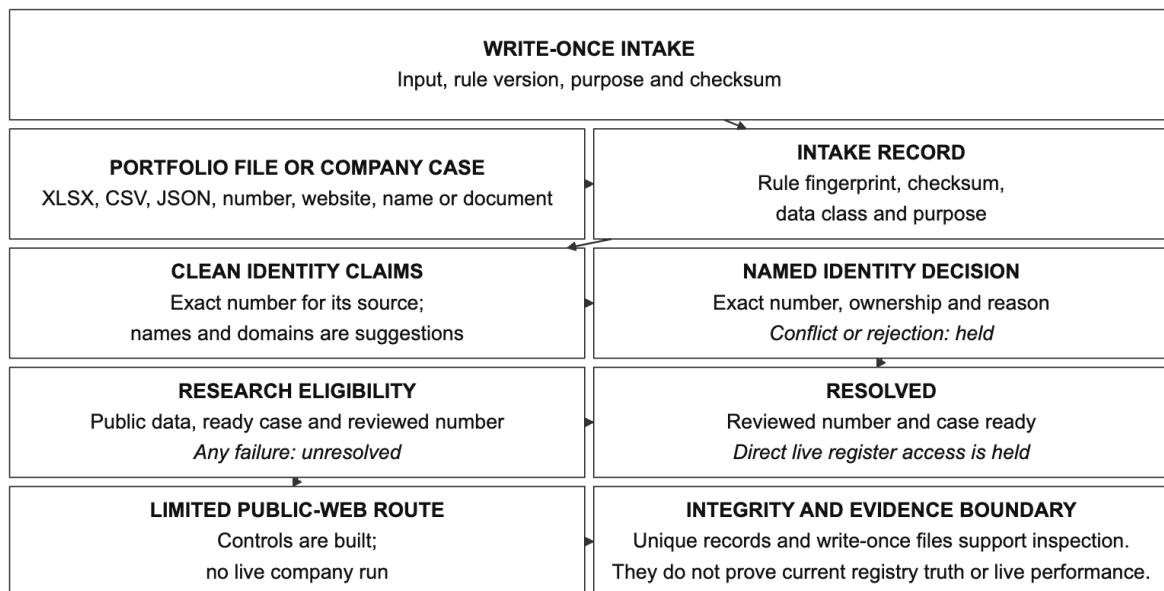


Figure 4.3. Legal-identity intake, decision and downstream research gate. Author synthesis of implemented local controls and held live-source boundaries; conceptual and non-empirical.

4.5 Fixed portfolio workflow and separate verification

The implemented portfolio workflow follows eight fixed stages: plan, resolve identity, collect evidence, extract values, normalise them, verify claims, compose a report and stop at human review. Each stage has a named functional role and a persisted run record with input and output hashes, status, timing, attempts and bounded metadata. It also preserves the order and outcome of every completed stage for later examination within this local release. An exception records failure and prevents later stages; unresolved identity also stops collection. These controls make execution inspectable, but artefact construction alone does not establish utility (Hevner et al., 2004; Pineau et al., 2021).

Collection stores run-linked evidence and its time decision before extraction. Only temporally eligible, trusted public evidence reaches the extraction provider; untrusted material is counted and bypassed. A strict schema binds extracted company, metric, period, value and exact evidence locator, while deterministic metric rules then normalise type, unit, currency and missing state. Existing run-evidence extractions are reused and provider attempts are stored. This supports traceable claim preparation, although attribution and repeatability remain distinct from correctness (Gao et al., 2023a;b; Pineau et al., 2021).

During verification, the workflow reviews sourceability, provenance, period, trust, value and currency before constructing a claim and linked verification record. The pure verifier returns only supported, contradicted, insufficient-evidence, stale or rejected-untrusted states; a current public conflict cannot be averaged away. Its role is separate from the extractor and composer, and it cannot silently rewrite their records. However, this is functional separation within one programme, not proof of independent human judgement or superior accuracy (Gao et al., 2023a;b; National Institute of Standards and Technology, 2023).

Composition places supported claims in the narrative and exposes other states as exceptions, then hashes the current, versioned report and moves it to pending review. A named reviewer may approve, reject or request an edit; each decision records actor, reason and report version. Editing creates a new section version, recalculates the content hash and revokes approval. Export remains unavailable until current approval, and optimistic locking plus a manifest-backed staged write guards stale or partial publication attempts (Amershi et al., 2019; Bućinca et al., 2021; Pineau et al., 2021).

The design favours deterministic orchestration, explicit stops and narrow roles over open-ended agent conversation. This reduces uncontrolled branching and makes failure locations auditable, but limits adaptive planning and does not show that more roles improve outcomes. Bounded attempts handle provider failure, while duplicate extraction and completed-export checks support local retry; they do not prove distributed exactly-once execution. Figure 4.4 compactly maps these control boundaries. Current code and synthetic tests establish the RQ1 implementation state and support the D0 verifier comparison in RQ2; human benefit remains a future pilot question (Guo et al., 2024; Peffers et al., 2007; Wu et al., 2024).

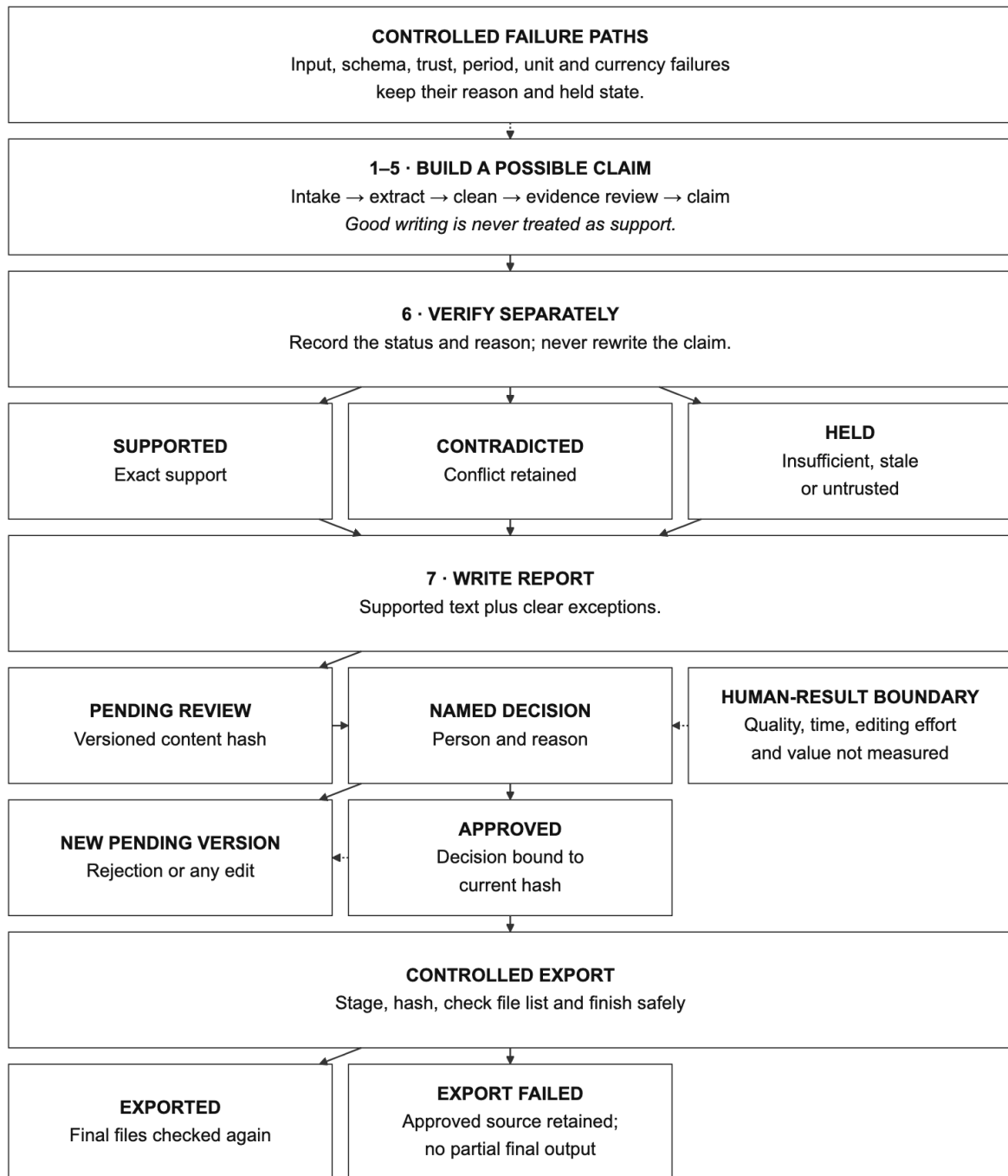


Figure 4.4. Fixed workflow, separate verification states, named approval and controlled export. Author synthesis of implemented local controls and held evaluation boundaries; conceptual and non-empirical.

4.6 Fixed source connectors, time and quality

The source layer is a versioned capability registry, not an unrestricted crawler. Each request specifies source, company, exact reviewed identifier, cutoff, fact keys and mode. Before adapter invocation, the registry rejects unsupported capabilities, unresolved, unreviewed or expired identity, and live retrieval with closed admission. A policy block is therefore pre-collection, not a persisted claim that the publisher held no record (Gebru et al., 2021; Pineau et al., 2021).

Only two deterministic adapters are usable, both through synthetic offline replay. Companies House manifest 1.4.0 matches an exact number and exposes company, filing and charge fields through JSON locators. UKRI manifest 1.3.0 matches an exact organisation identifier, selects the latest cutoff-available correction and emits stable project and outcome locators. Replays persist source version, bytes, checksum, retrieval and publication metadata, and a derivation hash. This supports repeatable inspection, not current registry truth or complete company coverage (Galanakis & Savagar, 2026; Hardman & Ramírez Santos, 2023; Pineau et al., 2021).

Time is claim-relative. Public evidence lacking recorded publication or availability time, published after the inclusive UK civil cutoff, future-effective, or expired before the period is ineligible; retrieval time stays separate. Post-cutoff events are omitted. A cumulative metric must exactly span stored programme start to cutoff. UKRI funding binds only when every in-window award has a finite explicit GBP amount; otherwise missing or non-GBP diagnostics and a missing state remain, without inferred zero or conversion (Hardman & Ramírez Santos, 2023; Kapoor & Narayanan, 2023; Nikiforova et al., 2020).

Returned facts require exact and structured locators, declared extraction method and schema, permitted metric, unit and currency, and valid time bounds. An exact lookup without a match is `no_record`; an unavailable source is `source_unavailable`; a policy block occurs before invocation; an invalid value is a typed missing state; and a terminal collection problem is `failed`. Separate storage or escalation explicitly prevents absence, invalidity and failure becoming interchangeable evidence (Gao et al., 2023a; Gebru et al., 2021; Nikiforova et al., 2020).

The versioned quality contract excludes untrusted or time-ineligible evidence, holds incomplete provenance, conflicts and terminal failures, and warns on bounded no-record, source-unavailable and expected-missingness cases. Table C.2 explicitly maps these decisions without treating a flag as truth. Deterministic replay makes identical inputs inspectable for later review, but coverage remains bounded by admitted snapshots and recorded source semantics. Direct live Companies House and UKRI retrieval remain held at G2; no live-source completeness or outcome claim follows (National Institute of Standards and Technology, 2023; Nikiforova et al., 2020; Pineau et al., 2021).

The detailed connector-state matrix is provided in Appendix C.

4.7 Bounded public-web company-research case study

The case-study path begins only after a public case has a reviewed Companies House number and a fixed reporting cutoff. The Responses API then uses `web_search` with `store=False` to plan discovery, but the application admits only canonical HTTPS URLs that also appear in the tool’s returned source list. Titles and model prose remain candidate metadata, not company facts for later controlled capture. This separation is necessary because sourced model output can still contain incomplete or incorrect support (Gao et al., 2023a; OpenAI, n.d.-b).

Each candidate is fetched by application code rather than trusted from search. It rejects credentials, non-default ports, local hosts and non-public DNS answers; checks robots; revalidates every redirect; permits six textual MIME types; and enforces elapsed, redirect and byte limits. The network backend connects to a validated public IP while the HTTPS pool retains hostname-based TLS verification before any extraction attempt. Retrieved pages are treated as hostile data because indirect prompt injection can enter through remotely supplied content (Greshake et al., 2023; National Institute of Standards and Technology, 2024).

Accepted bytes are converted to visible text, personal contacts are redacted, and snapshot files use content hashes, exclusive creation and owner-scoped directories. Extraction receives only bounded redacted text. A proposed statement survives only when its URL matches a captured source, statement equals evidence span, span occurs literally, cutoff and minimum-length rules pass, and dates or amounts appear in that span before any database admission. Thus, citations become inspectable evidence links rather than proof by presence alone (Gao et al., 2023a;b; National Institute of Standards and Technology, 2024).

The workflow persists four serial tasks: discover, capture, extract and compose, under a fingerprint covering identity, cutoff, model, prompt, policy and budgets. A later task cannot start before its prerequisite; retries are bounded; elapsed time, model calls, tool calls, tokens, source characters, bytes and redirects are counted. Named cancellation fences stale completion, while recovery records interrupted work and rejects changed contracts. These mechanisms make operational limits observable for each persisted attempt, consistent with documented monitoring and risk-response practice (National Institute of Standards and Technology, 2023; 2024).

Composition is deterministic over accepted claims and sources. It groups claims by category, marks multi-source disagreements for named resolution, lists blocked, unsupported, failed and uncovered categories, and hashes a pending-review profile. Figure 4.5 summarises this funnel. A missing source or unsupported claim therefore remains visible instead of completing a persuasive narrative. However, tests use fakes and controlled network responses: no live company, model or public-web run establishes coverage, accuracy, usefulness or superiority. This case study is implementation evidence within the stated evaluation rather than another empirical research question (Gao et al., 2023a;b; National Institute of Standards and Technology, 2023).

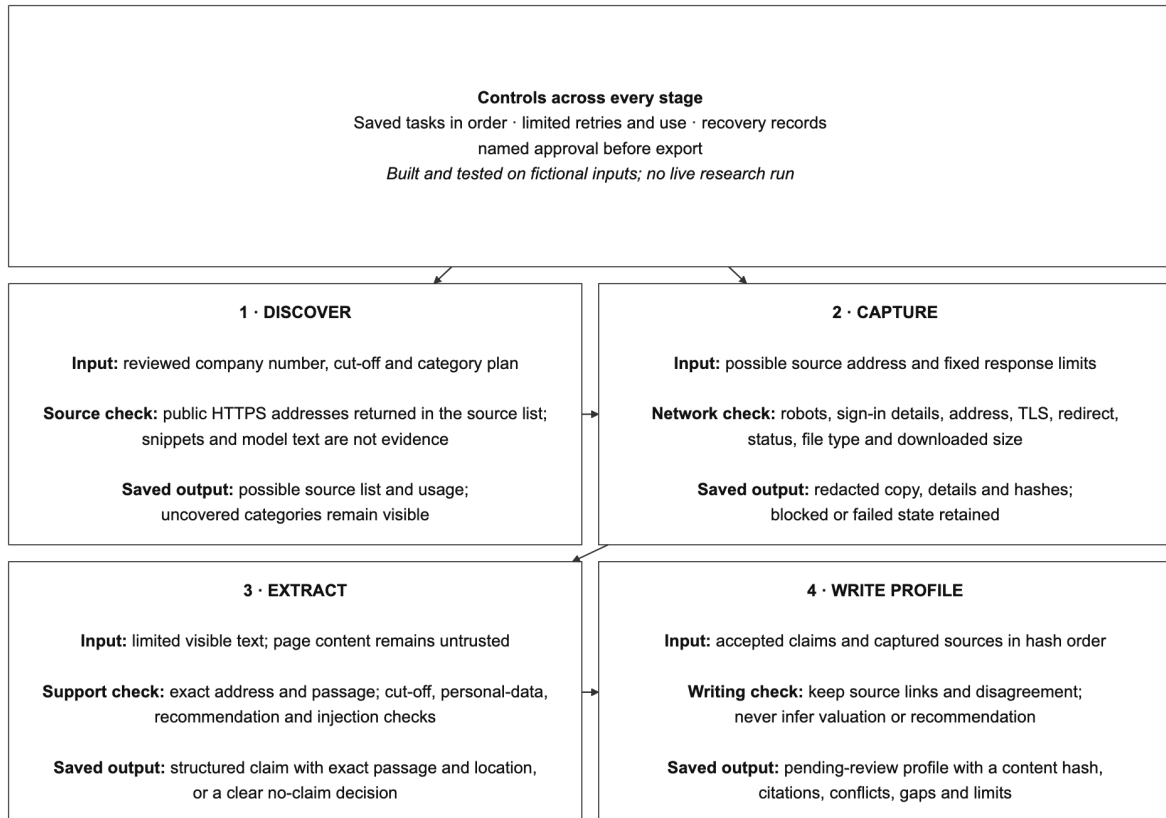


Figure 4.5. Limited public-web evidence process. Mermaid diagram developed by the author from the implemented controls and fictional tests. Live company research has not run.

Source note: OpenAI (n.d.-b, PDF pp. 1–3, 6) documents Responses web search and source URLs; Gao et al. (2023a, PDF pp. 1–3, 8–9) shows that citation quality and factual correctness require separate assessment. Greshake et al. (2023, PDF pp. 1–5, 25) and NIST (2024, PDF pp. 15–16, 44, 48) support treating downloaded content as untrusted and keeping risk controls, source records and human review. These sources do not certify this implementation or establish live performance.

4.8 Human review and controlled outputs

The portfolio-report path stops at pending review. Approval or rejection requires the configured reviewer identity, a substantive rationale and the current optimistic-lock token. Approval is permitted only when every claim has a recorded verification; an edit appends a section version, recalculates the report hash and revokes the prior decision. These controls create a deliberate decision point, but they do not demonstrate that review improves judgement (Amershi et al., 2019; Bućinca et al., 2021).

Company-research profiles use a separate, narrower gate. Before a named approve or reject decision, the service rechecks the profile hash, pinned run contract and completed composition task; stale locks, altered content and unfinished recovery are refused. Contradiction candidates and coverage gaps remain inside the reviewable dossier. Only approved profiles can be downloaded, as JSON or printable HTML, and neither format converts unresolved evidence into support (Gao et al., 2023a; National Institute of Standards and Technology, 2023).

For portfolio reports, export accepts only the current approved version and produces JSON, Markdown and accessible HTML. Files are staged below the local export root, bound to a versioned manifest and content hash, then reverified before reuse; a partial write is not marked exported. This is controlled local release, not autonomous publication. Native PDF or presentation export, remote distribution, production authentication and demonstrated reviewer effectiveness remain outside the implemented evidence (Gebru et al., 2021; National Institute of Standards and Technology, 2023; Pineau et al., 2021).

4.9 Failure, recovery, and implementation-status ledger

Failures keep separate names and remain visible. An invalid identity, policy rejection, unsupported request, unavailable source, no matching record, badly formed extraction, contradiction or old evidence cannot become a successful claim. Workflow errors stop later stages. Tasks that can be retried keep their attempts, limits and final reasons. Interrupted work can recover only under the same fixed run contract. A content change removes approval, and a staged export checks its file list before reuse. These controls support inspection and safe retry on one local system. They do not show recovery across several machines or continuous service availability (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

Table C.3 separates tested engineering work from work that is on hold. The main local workflow, source records, approval and export controls, offline connectors and gated public-web route have been built and tested with fictional or controlled inputs. Direct live official-source access, a live public-web check, the fixed company comparison, authorised human evaluation and production deployment have not been run, exist only as plans or are outside scope. This table prevents working code and a planned evaluation from being reported as evidence of effectiveness (Hevner et al., 2004; Peffers et al., 2007).

The detailed failure and implementation-status matrix is provided in Appendix C.

Evaluation and Results

5.1 Exact implementation and evidence snapshot

The evaluation record was created on 28 August 2026 using macOS Darwin 25.6.0 on Arm64 and Python 3.12.13. Git HEAD was 6356aa49cb3b, but the working tree contained dissertation and implementation changes, so the commit is not a complete identifier. Table G.1 records checksums for the dependency lock, protocol, test data and migrations; its text alternative retains the full values. This supports repetition without proving source truth or transferability (Gebru et al., 2021; Pineau et al., 2021).

Evidence comprises the local system at migration 0009, 286 automated tests and fourteen fictional D0 cases. Later interface and schema revisions described in Chapter 4 were not part of this dated snapshot and add no empirical result. D1 is planned, D2 sealed, and C0/C3 have no observations. The company-research route passed controlled tests but has not used a live company, model or public website. This chapter therefore reports engineering checks and designed mechanism cases, not human benefit, real-company coverage, investment outcomes or production readiness (Hevner et al., 2004; National Institute of Standards and Technology, 2023; Peffers et al., 2007). Conditions without observations remain null rather than estimated.

The complete implementation snapshot is provided in Appendix G.

5.2 Engineering validation

The final engineering run used the full automated test set rather than choosing only successful cases. All 286 tests passed and executed 85.58% of the source statements. Ruff found no lint errors, and strict MyPy found no type errors across 46 source files. A new SQLite database applied migrations 0001-0009, reported 0009 as the snapshot version and needed no further Alembic changes. These results support the internal consistency of the paths covered by the tests. Passing tests and code coverage cannot show the quality of real-world data, production security or practical value (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

The formatting check failed. Ruff found five files that it would change, made up of two source files and three test files. The dated snapshot retained those five findings; later code changes are not evaluated by this record. The test run also produced 22 warnings about features that Starlette and the SQLite date and time adapters may remove in future. Table G.2 therefore reports a mixed result instead of calling the whole run a pass. The formatting failure matters for repeatability, but it does not cancel the 286 passing tests (Gebru et al., 2021; Pineau et al., 2021).

Tests cover migrations, identity, time, source policy and records, verification, review, export, research tasks and unsafe-network controls using fictional data, fake providers and local files; no external source or model was contacted. Coverage of 85.58% identifies executed source statements, not complete requirements, branches or faults. Browser usability, concurrent production users, backup recovery and live sources were not tested. The result supports specified local mechanisms, not production readiness or investor effectiveness (Hevner et al., 2004; National Institute of Standards and Technology, 2023; Peffers et al., 2007).

The complete engineering-validation ledger is provided in Appendix G.

5.3 D0 fixture and comparison design

D0 contains fourteen labelled and fictional engineering cases, accepted through a versioned evaluation record. The cases cover complete and limited records, blank values compared with zero, mixed data types, conflicting and old evidence, and unsupported claims. They also cover unclear identity, repeated evidence, inaccessible sources, evidence in prose, mixed currencies and prompt injection. The data loader checks the file checksum, the test-only split rule and the `benchmark:d0` name. It then places case and evidence identifiers in that test namespace so the operational collector rejects them. This supports repeat testing without presenting fictional companies as real observations (Geburu et al., 2021; Pineau et al., 2021).

C1 and C2 use the same cases, data catalogue, cleaning rules and proposed values. C1 produces every eligible cleaned value unless the company identity is unclear. It keeps the available source details but has no separate verifier. C2 uses the same preparation and produces only values that a separate, fixed verifier marks as supported. Each condition ran three times. The first run was scored, and the complete set of outcomes was compared across repeats. Table 5.1 makes this one change between the conditions clear. Each comparison matches the same case in C1 and C2 (Guo et al., 2024; Wu et al., 2024).

Expected outputs and verification states share the visible development file that informed the rules. D0 therefore tests programmed gates on designed difficult cases, not independent real-company performance. It lacks event-link labels, a human reference, workload measures and variable model calls, so those measures are null or inapplicable. Final performance claims require a D1 pilot and sealed D2; any rule change after D2 opens makes the run exploratory to limit leakage (Kapoor & Narayanan, 2023; Peffers et al., 2007; Pineau et al., 2021).

Table 5.1. D0 fictional cases and C1–C2 comparison design

Design element	Shared input	C1 fixed baseline	C2 verification condition
Fixture	14 fictional labelled cases; manifest and fixture hashes; test-only benchmark:d0 names- pace	Same cases and order	Same cases and order
Edge-case coverage	Complete, sparse, blank/zero, mixed type, conflict, stale, un- supported, ambiguous identity, duplicate, inaccessible, narra- tive, mixed currency, prompt injection	Same labelled inputs	Same labelled inputs
Preparation	Catalogue, whether the source can be found, candidate value and currency, normaliser and expected states	Shared	Shared
Changed mechanism	The separately recorded verifi- cation step is the only planned difference	Emits eligible standardised candidates; unclear identity is held	Emits only candidates sup- ported by the verifier; unclear identity is held
Repeats and scoring	Three fixed repeats; the first run is scored; equal result lists show repeat consistency	Three repeats	Three repeats
What cannot be con- cluded	The visible development la- bels and rules were designed together	No independent or real- company baseline estimate	No estimate of the verification effect on new cases

Source note: Author summary of the runnable evaluation list, D0 cases, evaluator and fixed study plan. The table shows the implemented difference between the conditions. It does not prove that one condition is better in real use or that people would benefit.

5.4 D0 synthetic mechanism results

The repeated run produced fourteen scored cases for each automated condition. C1 produced eleven claims: five correct positive claims and six incorrect positive claims, with no required claim missing. C2 produced five claims, all correct, and returned no claim for the other nine cases as intended. C1 therefore had precision of 0.455, recall of 1.000 and F1 of 0.625. C2 recorded 1.000 for all three measures. This apparent improvement applies only to the designed D0 cases and comes from blocking the six candidates designed to be unsupported (Gao et al., 2023a;b).

The unsupported-claim rate fell from 0.545 in C1 to 0.000 in C2, while the supported-claim rate rose from 0.455 to 1.000. Complete source records rose from 0.818 to 1.000 because two C1 claims did not have every required source detail and C2 did not produce them. All three rates use the number of claims produced as their total, and that total falls from eleven to five. The result therefore shows more selective output, not wider source discovery or a more complete report (Gao et al., 2023a; Gebru et al., 2021).

Both conditions followed the schema correctly, cleaned every value correctly and produced exactly the same outcomes across the three repeats, giving 1.000 on each measure. C1 was correct in 0.571 of verification and no-claim decisions, compared with 1.000 for C2. C1 missed both labelled contradictions and C2 found both, so their contradiction scores were 0.000 and 1.000. This result is based on just two contradictions. Table 5.2 keeps that warning, and Figure 5.1 shows the main rates without combining them into one score (Nikiforova et al., 2020; Pineau et al., 2021).

Mean time was recorded as zero milliseconds for both conditions because the evaluator cuts times below one millisecond down to the nearest whole number. It cannot support a speed comparison. The recorded LLM cost was USD 0 because both routes are fixed and run locally. Measures for identity, extraction, timing of evidence, source quality, event links, report correctness and reviewer value were blank because D0 has no suitable labels or observations. Keeping a blank result instead of changing it to zero avoids suggesting a measured failure or success (Gebru et al., 2021; National Institute of Standards and Technology, 2023).

Looking at individual cases explains the overall change. C1 incorrectly produced claims for the conflict, old evidence, unsupported evidence, inaccessible source, mixed currency and prompt injection cases. C2 gave them cautious unsupported states and produced no claim. It kept the complete case, the true zero, mixed data types, matching repeated evidence and claims supported by prose. This shows that the intended gates worked for the categories included in D0. However, each category has only one case, so the result cannot describe a wider distribution or show that the system is robust beyond these examples (Gao et al., 2023b; National Institute of Standards and Technology, 2023).

The verifier rejected every deliberately unsupported, conflicting, old, inaccessible, mixed-currency or prompt-injection candidate in D0 and kept all five claims designed to be supported. This does not show that the same rules will find, understand or resolve unexpected evidence from real companies. C2's perfect values need particular caution because the test labels and fixed rules were developed together. A D1 pilot and a fixed D2 comparison labelled independently are needed before estimating performance on new cases, uncertainty or practical value (Hevner et al., 2004; Kapoor & Narayanan, 2023; Peffers et al., 2007).

Table 5.2. D0 fictional-case results by automated condition

Measure	C1 baseline	C2 verifier	Total or meaning
Cases; emitted claims	14; 11	14; 5	The same fourteen paired fictional cases.
Confusion counts (TP / FP / FN / TN)	5 / 6 / 0 / 3	5 / 0 / 0 / 9	Expected-emission labels in visible D0 fixture.
Precision / recall / F1	0.455 / 1.000 / 0.625	1.000 / 1.000 / 1.000	Emitted and expected claims; evidence from designed cases only.
Unsupported / support rate	0.545 / 0.455	0.000 / 1.000	Each uses emitted claims: 11 for C1, 5 for C2.
Source-record completeness	0.818	1.000	Complete source records among emitted claims.
Schema / normalisation accuracy	1.000 / 1.000	1.000 / 1.000	Fourteen labelled cases per condition.
Verification / abstention accuracy	0.571 / 0.571	1.000 / 1.000	Fourteen labelled cases per condition.
Contradiction accuracy	0.000	1.000	Two applicable conflict cases only.
Exact repeat consistency	1.000	1.000	Three fixed result lists.
Mean duration; LLM cost	0 ms; USD 0	0 ms; USD 0	Values below one millisecond are shown as zero; both routes were local and fixed.
Other outcome layers	NULL		No applicable identity, extraction, temporal, quality, event, report or reviewer labels.

Source note: Fresh three-repeat D0 evaluator output, SHA-256 e3be8f62...cdda27 (full value retained in the text alternative). Values come from fictional cases. They do not support a confidence interval or a claim about performance in real use.

Table 5.3. Examples of accepted and held D0 claims

Case and proposed claim	Evidence	Verification rule	Decision	Report consequence
Observed zero: jobs created = 0	Portfolio submission records 0 for SYN-PERIOD; locator: submission synthetic B2	Zero is a value when the typed observation and evidence agree	Supported	Claim is emitted with its source record
Contradiction: grant funding = GBP 100	Synthetic register records GBP 200 for the same period; locator: fixture conflict	Candidate value and currency must agree with eligible evidence	Contradicted	Claim is withheld and the conflict is retained
Stale period: awards received = 1	Matching value exists only for PRIOR-PERIOD; locator: fixture stale	Evidence period must match the report period	Stale	Claim is withheld; the old evidence remains visible
Prompt injection: new partnerships = 5	Matching value appears in the fixture injection record, but it is marked untrusted	Instruction-like untrusted evidence cannot support a claim	Rejected untrusted	Claim is withheld and no instruction is followed

Source note: Exact fields from `fixtures/evaluation_cases.json`. All values, publishers and locators are fictional and were created for the D0 engineering evaluation.

Measure	C1: no verifier	C2: verifier
Precision	0.455	1.000
F1	0.625	1.000
Claim-support rate	0.455	1.000
Complete source records	0.818	1.000
Unsupported-claim rate ↓	0.545	0.000

Figure 5.1. D0 metric profile for C1 and C2. Author Mermaid visualisation of the current D0 output from fourteen visible synthetic cases. Emitted-claim denominators differ (C1 eleven; C2 five), so the values are neither coverage nor utility scores and are non-empirical beyond the constructed fixture.

5.5 Company-research contract and adversarial results

A focused run passed 37 company-research, migration and fixture tests. The controlled end-to-end case resolved a reviewed company number, admitted bounded search suggestions and content, accepted exact supporting passages, composed profile version three and required named review before download. Other cases covered run fingerprints, replay, budgets, cancellation, recovery, source ownership, contradictions and stale approval. These local rules worked under test; live-source coverage and profile accuracy remain unmeasured (Gao et al., 2023a; National Institute of Standards and Technology, 2023).

Hostile-input tests rejected credential-bearing or non-HTTPS URLs, blocked ports, private or changed network addresses, excessive redirects, robots exclusions, wrong file types and oversized responses. Controlled components checked connection to a public IP with HTTPS hostname verification. Downloaded text remained untrusted; suggested URLs had to appear in the bounded source list, retries consumed the run budget, and exact captured passages supported accepted values. These selected cases do not establish complete SSRF or prompt-injection protection (Greshake et al., 2023; National Institute of Standards and Technology, 2024).

Table F.3 maps each control group to residual risk. Tests also covered publisher diversity, duplicate snapshots, identity passages, timeout failure, zero redirects and run-specific cleanup. Inputs were local fixtures, fake providers or controlled responses, with one Starlette deprecation warning. No OpenAI, Companies House, publisher or open website call occurred; delay, tokens, cost, paywalls, DNS change, source availability, open-web coverage and profile usefulness remain unmeasured (National Institute of Standards and Technology, 2023; OpenAI, n.d.-b; Pineau et al., 2021).

These security tests check stated examples rather than measuring a full attack campaign. A pass means that one specified input was refused or kept as designed. It gives no rate for incorrect blocks, missed attacks, range of attacks covered or statistical uncertainty. Repeated tests that generate many inputs, checks of dependencies and penetration tests designed by an independent person would be needed to estimate how well the system resists attacks beyond the current examples (Greshake et al., 2023; National Institute of Standards and Technology, 2023).

The detailed adversarial-control matrix is provided in Appendix F.

5.6 Held future public-company source-discovery comparison

This comparison did not run and provides no result. A future study would give approved official connectors and bounded assisted discovery the same companies, reporting cut-offs, source window and output rules. Search results would remain candidate sources until admitted and matched to an exact passage (Gao et al., 2023a; OpenAI, n.d.-b). The company set, reference labels, prompts, budgets, source rules and stops must be frozen before collection to prevent unequal information access (Kapoor & Narayanan, 2023; Pineau et al., 2021). The held protocol and null result fields are in Appendix F.

5.7 Manual and human-in-the-loop comparison

C0 manual work and C3 participant review did not run and are not active research questions. The interface records versioned edits and decisions, but there are no observations of correctness, time, cost, editing or usefulness. Human control can support correction while also creating over-reliance or burden (Amershi et al., 2019; Buçinca et al., 2021); those effects require the prospective pilot in Section 6.7.

The original comparison status is retained in Appendix F as protocol history.

5.8 Negative, null, failed, and unavailable outcomes

Negative results were kept. Ruff’s formatting check failed for five files, and the passing test set produced 22 warnings about future changes. C1 produced six incorrect positive claims in D0 and missed the only conflict. Timings of zero milliseconds gave no useful information because the measure cut times below one millisecond down to a whole number. These results were not removed from the totals or presented as successful checks. They show the engineering and measurement work that remains (National Institute of Standards and Technology, 2023; Pineau et al., 2021). The formatting failure is still present in the evaluated record.

Blank and unavailable outcomes were retained. D0 lacks identity, extraction, evidence-time, source, event, report and reviewer labels; C0/C3 have no observations. D1 is planned, D2 sealed, live official access held, and public-web, fixed-company, participant and production runs absent. Blocked request, unavailable source, no record, unsupported claim, contradiction and failure remain distinct. The chapter supports local-mechanism claims only, not real-company performance, human benefit, complete coverage or investment outcomes (Gao et al., 2023b; Gebru et al., 2021; Hevner et al., 2004).

Discussion

6.1 RQ1: Evidence preservation in the implemented workflow

RQ1 asks how the implemented workflow preserves identity, time, missing states, provenance and approval boundaries from input to report. The repository and engineering checks show that these controls exist in one local workflow. Inputs retain dataset and period identifiers; company matches can enter an explicit hold; missing values keep typed reasons; source records retain locators and checksums; verifier decisions are persisted; and export requires approval of the current report version. This is evidence that the workflow is inspectable, not evidence that it is ready for production (Hevner et al., 2004; Peffers et al., 2007).

D0 supplies a narrower behavioural result. Both conditions normalised all fourteen fictional cases as expected and returned the same ordered outcomes in three repeats. C2 emitted the five supported cases and held the remaining nine. The accepted observed-zero case shows why a typed zero must not be collapsed into missing data. The contradicted, stale and untrusted examples show how period, value and trust rules change the report consequence. These decisions remain traceable to exact fixture records (Gao et al., 2023a;b).

Traceability does not make a source true. A fixed workflow can reproduce an incorrect publisher statement, and an exact passage can support a statement that is still misleading. The contribution is therefore the ability to inspect origin, transformation, support decision and release authority. Independent source judgement and coverage assessment remain separate tasks (Gao et al., 2023b; Pineau et al., 2021).

RQ1 is answered for the implemented local mechanisms and controlled inputs. It establishes a complete record of why a claim was emitted or withheld within this prototype. It does not estimate accuracy across real portfolios. Chapter 7 gives the full validity boundary (Gebru et al., 2021; National Institute of Standards and Technology, 2023; Pineau et al., 2021).

6.2 RQ2: Observed effect of separate verification

RQ2 compares C1 and C2 on identical D0 candidates and evidence. C1 emitted eleven claims, including six designed to be unsupported. Its precision was 0.455 and its unsupported-claim rate was 0.545. C2 emitted five claims, all labelled as supported, giving precision of 1.000 and an unsupported-claim rate of 0.000. Recall remained 1.000 in both conditions. The observed effect was selective admission, not discovery of additional evidence (Gao et al., 2023a;b).

The lower claim count is important. C2 did not improve by producing a more complete report; it withheld candidates for conflicting values, a stale period, absent or inaccessible support, mixed currency and untrusted content. This was correct for D0, but a similar gate could falsely reject a valid claim in new data. D0 cannot estimate false rejection outside its visible designed cases (Kapoor & Narayanan, 2023; National Institute of Standards and Technology, 2023).

The result supports a separately recorded verification decision in this workflow. It does not show that multi-agent systems are generally superior. C1 and C2 share one program, catalogue and normalisation route. A deterministic pipeline or single grounded agent could apply the same support rule. The experiment isolates the gate and its authority over composition, not the effect of a persona, conversation style or number of agents (Guo et al., 2024; Wu et al., 2024).

No useful speed or cost comparison was produced. Both routes used local fixed rules, model cost was zero and sub-millisecond times were truncated to zero. Role separation also adds states, hand-offs, storage, tests and maintenance. A later pilot must measure these costs alongside false acceptance, false rejection and reviewer work (Hevner et al., 2004; Pineau et al., 2021).

6.3 Why functional role separation was selected

The selected design separates production from verification and keeps approval outside both. The planner and collectors can propose work, but they cannot approve a report. The verifier can accept, hold or contradict a candidate, but it cannot rewrite the claim to make it pass. The composer can use only admitted claims, and the reviewer decides whether the current version may be exported. Persisted hand-offs make each decision recoverable and allow a failure to stop one stage without silently changing another (Guo et al., 2024; Wu et al., 2024).

This design also supports narrower permissions. Public discovery, protected collection, local restricted-data processing and final approval do not share one unrestricted tool boundary. The trade-off is more code and more operational state. Multi-agent trace studies report coordination and task- verification failures, so added roles are themselves a source of risk (Cemri et al., 2025; National Institute of Standards and Technology, 2023). The prototype uses a fixed state machine rather than an open-ended conversation to keep this cost visible.

The alternatives in Table 2.4 remain credible. Manual work can use contextual judgement; a deterministic pipeline can be simpler for stable structured inputs; and a single grounded agent can reduce hand-offs. The selected workflow was useful for demonstrating explicit support and approval boundaries. The evaluation does not prove that it is the best option for every reporting team (Gao et al., 2023c; Guo et al., 2024).

6.4 Public-web case study and future comparator

The public-web route extends the implementation from structured replay to bounded source discovery. Web search proposes URLs, while local controls decide whether a page may be collected and used. A claim still requires a saved source, exact supporting passage, source decision and approval. This separation follows the distinction between finding material and establishing claim support (Gao et al., 2023a; OpenAI, n.d.-b).

Controlled tests covered reviewed identity, HTTPS and network checks, request limits, write-once captures, personal-contact redaction, exact passages, contradictions and approval. They used fake

providers, fixtures and controlled responses. They establish behaviour for those represented cases, not live source coverage or attack resistance across the open web (Greshake et al., 2023; National Institute of Standards and Technology, 2023).

One future managed-platform comparator is Grok Bot. Its official documentation describes persistent named Bots with defined jobs, tools and sources, working context, hand-offs and approval boundaries. This could replace part of the custom orchestration and interaction layer, but lower setup or maintenance effort is a hypothesis to test, not a current finding. The comparison should ask whether Grok Bot can implement an equivalent evidence, verification and approval contract. It must use the same inputs, source windows, output requirements and scoring rules. Measures should include setup effort, active analyst time, supported-claim precision, evidence coverage, false acceptance, false rejection, traceability, approval compliance, failure recovery and total cost. Product-managed Bots do not automatically replace source admission or claim verification (Dibia et al., 2024; Peffers et al., 2007; Pineau et al., 2021; SpaceXAI, 2026).

6.5 Relationship to literature and design trade-offs

The findings agree with literature that separates factual correctness, citation support and source attribution. Exact locators and saved copies make a decision inspectable, but they do not establish source truth or complete coverage (Gao et al., 2023a;b). The observed D0 change is consistent with the purpose of a verification gate because the fixture was built to test that gate. It is not an independent confirmation of a general theory.

Fixed missing-state, cut-off and version rules also support repeatability. Their weakness is that a wrong or outdated rule repeats consistently. The rules therefore need named ownership, change control and re-evaluation when sources, schemas or reporting needs change (Gebru et al., 2021; Nikiforova et al., 2020; Pineau et al., 2021).

Human approval provides accountability for release, not automatic correctness. Interfaces should show uncertainty and allow correction, while studies of reliance show that people may still follow incorrect advice (Amershi et al., 2019; Bućinca et al., 2021). The pilot must measure reviewer edits, decisions and time instead of treating an approval button as evidence of benefit.

6.6 Generalisability and practical interpretation

The result applies to one local Python, FastAPI, SQLite and Next.js prototype, fourteen designed D0 cases and controlled security tests. It does not estimate behaviour across sectors, document types, reporting periods, reviewers or changing sources. Early-stage private-market evidence is especially incomplete and definition-sensitive (British Business Bank, 2025; Kaplan & Lerner, 2016). Statistical generalisation would require an independently selected and labelled set (Kapoor & Narayanan, 2023; Pineau et al., 2021).

Transfer to another setting requires a new data dictionary, legal identity source, source licences, missing-value rules, reporting cut-off, evidence contract, security boundary and approval owner. Checksums and versions explain differences; they do not keep performance constant when sources or models change (Gebru et al., 2021; National Institute of Standards and Technology, 2023). The detailed transfer checklist is in Appendix H.

The prototype is therefore best understood as a testable reporting control system. It does not rank

companies, predict performance or provide investment advice. Practical usefulness remains a pilot hypothesis (Financial Conduct Authority, 2024; Hevner et al., 2004).

6.7 Prospective business pilot

The proposed pilot is a time-limited evaluation of the existing portfolio XLSX import as the named input integration. The prototype's Next.js review interface is the downstream report interface, with versioned JSON and accessible HTML exported only after approval. The pilot owner approves scope and the go/no-go decision. A reporting analyst prepares the manual baseline and operates each reporting case. A technical operator manages the local service, versions and failures. A reviewer or approver checks the evidence and current report version. A data and security owner approves data classes, source licences, storage, retention and any external-model route (Amershi et al., 2019; National Institute of Standards and Technology, 2023).

Before data collection, the team freezes the companies, reporting periods, eligible sources, schema, reference decisions, task completion rule and acceptance thresholds. It records setup effort in staff-hours for data mapping, source admission, environment configuration, training and dry runs. For every report it records analyst active minutes, reviewer active minutes, elapsed time, review and rework time, retries, model or API tokens, provider charges, source-licensing charges, storage and security costs. Cost per report is the allocated setup cost plus staff time, API use, licences, storage and security, divided by completed reports. No saving is assumed before this matched baseline exists (Peffer et al., 2007; Pineau et al., 2021).

Quality measures are supported-claim precision, evidence coverage, false acceptance, false rejection, reviewer edits, completion rate and source failures. Operational measures are active time, elapsed time and cost per report. The manual baseline uses the same inputs, source window and completion rule. Results are paired by company-period, and failed or abandoned cases remain in the denominator where the measure requires them. Data-quality measures must remain tied to the intended reporting use (Nikiforova et al., 2020), and reviewer disagreement must not be treated as proof that either person is correct (Artstein & Poesio, 2008).

The pilot proceeds only if authority, source licences, reference labels, storage controls, trained owners and rollback are in place. Before seeing outcomes, the pilot owner and reviewer set the minimum acceptable quality and completion values and the maximum acceptable false-acceptance, false-rejection and cost values. The recommendation is go only when those frozen criteria are met, approval cannot be bypassed, every exported claim has the required evidence record, and no stop event occurs. It is no-go or redesign when a criterion fails. This avoids inventing numeric targets in the dissertation while leaving a checkable decision rule (National Institute of Standards and Technology, 2023; Peffer et al., 2007).

The run stops immediately for unclear authority or consent, unexpected restricted or personal data, credential exposure, a source-licence or robots breach, external transmission outside the approved route, a broken evaluation freeze, evidence-record loss, approval bypass, uncontrolled cost or a security incident. The event, affected cases and recovery decision are recorded. Expected time or cost reduction remains a hypothesis until the paired pilot demonstrates it (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

Ethics, Governance and Limitations

7.1 Data classification, privacy and purpose limits

At intake, evidence is marked restricted, internal, public or synthetic. Restricted portfolio and internal operating data stay local and cannot reach an external model. External processing of public or synthetic material requires explicit selection and a stated case purpose. Data minimisation reduces exposure, but classification depends on human accuracy and establishes neither a lawful basis nor university approval (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

Local source copies and exports are checksum-linked, write-once where required, excluded from Git and privately permissioned. Personal contacts are removed before accepted public pages are stored. A checksum detects file change; it does not anonymise or encrypt data. Public availability does not remove privacy, licence or purpose concerns, and backups, temporary storage or cloud-synchronised folders require separate review (Central Digital and Data Office, 2023; Gebru et al., 2021).

Real restricted data and participant work remain held pending an approved data-management and ethics plan identifying control, purpose, access, retention, deletion and incident ownership. No retention period is invented. The prototype demonstrates local purpose limits, not data-protection compliance or permission to reuse information about portfolios, employees, contacts or reviewers in a future service (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

7.2 External-model processing and retention

The optional external route sends only approved public or synthetic company-level material and a reference that does not expose evidence. It requires open collection/model gates, reviewed identity, named reviewer, approved route and a runtime key. Requests use `store=False`, strict schemas and fixed limits. Restricted, internal or personal-contact content is prohibited, and model output cannot approve or export a report (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

These controls limit transmission but cannot guarantee provider deletion. OpenAI distinguishes saved application state from abuse-monitoring retention; `store=False` is not Zero Data Retention. Before live use, account settings, eligibility, current behaviour, terms and data location require verification. No live request occurred, so handling, tokens and retention status were unobserved; a future run must record the setting and confirming evidence (Central Digital and Data Office, 2023; OpenAI, n.d.-a).

7.3 Security and failure threats

Public pages may contain hidden instructions, misleading identities or malformed content. The route treats downloaded text as data, limits model tools, rejects instruction-like passages and requires exact support before claim admission. Tests cover selected risks; the controls cannot recognise every hostile form or guarantee that a plausible supported statement is harmless (Greshake et al., 2023; National Institute of Standards and Technology, 2024).

Collection checks HTTPS, embedded credentials, ports, network addresses, redirects, robots, type, time and decompressed size. It connects to a checked public IP while verifying the hostname. Controlled internal-service, address-change, redirect, type and size tests passed. Hosting would still require egress isolation, authentication and tenant separation. These local tests verify selected rules, not a production boundary, independent penetration test or estimate of residual security risk (National Institute of Standards and Technology, 2023; 2024).

Budgets, bounded retries, cancellation, task fingerprints, checksums and staged export reduce uncontrolled cost, stale completion, changed files and partial failure. A forced stop may still require deliberate recovery, while dependencies or sources can change. Recorded failure states support inspection but do not establish service availability, disaster recovery or attack coverage (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

7.4 Investment and public-person boundaries

The report links company events to evidence; it does not advise buying or selling, rank companies, create speculative valuations or assess suitability. Funding, awards, contracts, filings and attention are contextual and do not themselves cause or measure quality or future performance. Because financial information can influence decisions, external use requires separate review of purpose, audience and presentation (Estrin et al., 2024; Financial Conduct Authority, 2024).

The research concerns companies, not individuals. Captured contact details are removed; director, employee, investor or journalist profiles, inferred sensitive traits and reputation scores are excluded. A named public statement may support a relevant company claim but cannot become a judgement about that person. Human approval cannot expand these limits (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

7.5 Research and operational limitations

The principal evidential limit is D0: fourteen fictional development cases built around known rules, so the verifier difference may reflect joint case and rule design. The original manual, participant, live-web and managed-platform comparisons were removed from the active research questions because their required observations were unavailable. The study cannot estimate new-company accuracy, parity with manual work, human benefit or population performance (Kapoor & Narayanan, 2023; Pineau et al., 2021).

Sources, licences, data formats, publisher pages, search order, model behaviour and provider policy can change. Checksums detect saved-input changes but cannot preserve access or meaning. The researcher designed the system and cases and performed scoring and interpretation, creating confirmation risk. The final reference therefore requires independent construction and pre-study evaluation freeze (Gebru

et al., 2021; National Institute of Standards and Technology, 2023).

Operationally, the prototype is local, single-user, serial and SQLite-based. It lacks production authentication, tenant separation, egress isolation, central monitoring, backups, service targets and tested disaster recovery. The PDF had visual checks but no independent accessibility or assistive-technology study. The dated snapshot recorded five formatting failures and 22 warnings; later changes remain unevaluated. These limits do not erase the controlled D0 result, but they bar usability, resilience, maintainability and production-readiness claims. Table H.3 summarises residual authority and risk (Amershi et al., 2019; Hevner et al., 2004; Peffers et al., 2007).

The detailed residual-risk matrix is provided in Appendix H.

Conclusion and Future Work

8.1 Answers to the primary question, RQ1 and RQ2

The primary question is answered within the controlled scope. The implemented workflow can preserve identity, reporting-period, missing-state, provenance, verification and approval records from intake to report export. Engineering checks and repository artefacts establish that these boundaries exist in the evaluated local version. They do not establish operational usefulness or production readiness (Hevner et al., 2004; Peffers et al., 2007).

For RQ1, the strongest result is traceable execution. D0 decisions can be followed from a fictional candidate and its evidence to the verifier state and report consequence, and the complete outcome list was identical across three repeats. The observed-zero, contradiction, stale-period and untrusted examples show the practical meaning of the evidence rules. This answers how the implemented controls preserve reporting meaning within D0 (Gao et al., 2023a;b; Pineau et al., 2021).

For RQ2, separate verification changed claim admission under equal inputs. C1 emitted six candidates that D0 labelled unsupported, while C2 held all six and retained the five supported claims. Precision changed from 0.455 to 1.000, the unsupported-claim rate from 0.545 to 0.000, and recall remained 1.000. This is an observed verification-gate effect on fourteen designed cases. It is not evidence that multiple agents are generally superior or that the same rates will hold for real companies (Guo et al., 2024; Kapoor & Narayanan, 2023).

The dissertation’s conclusion is therefore narrower and stronger than the original research contract. It demonstrates an evidence-first implementation and one controlled verification effect. Manual comparison, participant review, live source use and business benefit are outside the active questions because the required observations were unavailable. Their absence does not make the executed result incomplete; it sets the boundary for future evaluation (Hevner et al., 2004; Pineau et al., 2021).

8.2 Demonstrated contribution

The main contribution is a reporting workflow that makes source admission, claim support, conflicts, missing information and release authority explicit. Functional role separation gives production, verification and approval different permissions and persisted hand-offs. The design is inspectable because the saved records link each decision to a source, rule, version and output (Buneman et al., 2001; Pineau et al., 2021).

The empirical contribution is the paired D0 comparison. It reports aggregate metrics and exact case

examples, including a correct accepted zero and three different reasons for holding a claim. The result supports the verification contract on the fixture rather than a new algorithm, a persona effect or a claim of production performance (Gao et al., 2023a;b).

The practical contribution is an executable pilot design. It names the existing XLSX input and Next.js review interface, assigns staff responsibilities, defines cost and time records, specifies quality and operational measures, and sets go/no-go and stop rules. Time saving and cost reduction remain hypotheses until that pilot is authorised and run (National Institute of Standards and Technology, 2023; Peffers et al., 2007).

8.3 Sequenced future evidence

The first next step is the prospective business pilot in Section 6.7. It should freeze the data, source window, reference decisions, completion rule and acceptance criteria before collecting paired manual and workflow observations. The pilot must record active and elapsed time, reviewer work, supported-claim precision, evidence coverage, false acceptance, false rejection, source failures, completion and cost per report. A pilot result should not be used to tune a sealed final evaluation set (Kapoor & Narayanan, 2023; Pineau et al., 2021).

A separate live-source study can then compare approved official connectors with bounded assisted discovery using identical companies, dates, source limits and output requirements. Any external model route requires current provider controls, approved data classes and recorded usage. An independent accessibility and security assessment is also needed before remote or multi-user operation is considered (Central Digital and Data Office, 2023; National Institute of Standards and Technology, 2023).

Grok Bot is a named future implementation comparator, not a presumed replacement. Its documented Bots can hold defined jobs, working context and approval boundaries, but this dissertation did not implement or test them. A future study should ask whether Grok Bot can reproduce the same evidence-admission, verification and approval contract with less setup and maintenance effort. Equivalent inputs, source windows and output requirements are required. The comparison should measure setup effort, analyst time, supported-claim precision, evidence coverage, false acceptance, false rejection, traceability, approval compliance, recovery and total cost. It must keep orchestration convenience separate from evidence quality and reporting correctness (Dibia et al., 2024; Guo et al., 2024; Pineau et al., 2021; SpaceXAI, 2026).

The full evidence roadmap is provided in Appendix H. Each future stage requires an explicit authority and stop decision. None is presented as a result of this dissertation (National Institute of Standards and Technology, 2023; Pineau et al., 2021).

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Evidence of Required Ethics Training

Ethical Approval or Waiver Confirmation

Requirements and Implementation Traceability

Table C.1. Prototype requirements, controls and limits

Requirement	Person, data and action in scope	Control in the prototype	Work still needed	Outside scope or reason to stop	Evidence and status
Local runtime and accountable actor NFR-SEC-001; FR-UI-003-004	One configured local operator uses the loopback Next.js dashboard, private FastAPI service, SQLite and ignored local files.	Next.js is the only published browser surface; its same-origin server-side proxy reaches FastAPI privately. Host/client checks, server-held CSRF credentials and configured reviewer identity protect changes.	Formal accessibility/security review and any wider deployment design require separate evidence and authority.	No anonymous, remote or multi-user access. Reject non-loopback clients, unexpected hosts, missing CSRF or missing configured reviewer.	<i>web.py</i> ; dashboard proxy; Compose and route tests. IN PLACE Local only; not production sign-in.
Identity-first intake FR-ID-001-004; FR-INT-001, 006	A submitted Companies House number is an identity claim; a named reviewer may accept or reject it with rationale.	A number with a valid format creates an unresolved company and unreviewed identifier. Names and domains do not merge records automatically. Identity conflicts and data-class mismatches stop the case.	Before direct live registry use: close G2 with current interface/terms, credential, identifier, storage, cutoff and attribution review.	Syntax is not legal-identity authority. Do not collect or link facts while identity is unresolved, conflicting, rejected or from another classification boundary.	Intake service; ADR-0007; tests. IN PLACE Offline; live registry held.
Data class and authority NFR-SEC-002-005; NFR-RES-001	Restricted and internal material stays local. Accepted public evidence and fictional cases follow separate routes.	Fixed data classes; purpose and person records; write-once files with private “0600” permissions; records without content; provider check.	Verify lawful authority, controller, access, retention and deletion, and any ethics/consent terms before restricted-data or participant work.	Never place restricted or internal content in Git, fixtures, raw logs or an external model. Unclear authority or unexpected personal data stops processing.	<i>enums.py</i> ; <i>security.py</i> ; boundary tests. CONTROL IMPLEMENTED ; authority not verified.
Public and model boundary FR-EXT-004-008; FR-RES-001-004	Only a public case with a reviewed company number and cut-off may enter limited web search, collection and optional extraction. Fictional cases test the route offline.	External and collection gates start closed. Snippets and model text are suggestions, not evidence. Protected HTTPS collection, rejection of hidden instructions, “store=False”, strict passage and schema checks, and separate verification are required.	Separately authorise and evaluate the live public company path; finish source-specific G2 review and verify provider controls at execution. Direct registry execution remains P1.	No restricted or internal model input, person profiling, arbitrary browsing, snippet support, autonomous recommendation or model approval/export. Missing, held or conflicting evidence cannot become support.	<i>bootstrap.py</i> ; ADR-0008; controlled boundary tests. CODE AND LOCAL TEST EVIDENCE ; live company route not run.

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Table C.1 continued: prototype requirements, controls and limits

Requirement	Person, data and action in scope	Control in the prototype	Work still needed	Outside scope or reason to stop	Evidence and status
Review and export FR-VER-002; FR-REP-004–009	The configured reviewer assesses a versioned report or profile with verification records, then approves, rejects or returns it.	Person, reason and version record; version lock; edits remove approval; no export before approval; files checked against a list and hashes before reuse.	Authorised C0/C3 observation and reviewer utility remain pending; production roles or signed approval need a separately scoped design.	No model or workflow stage can approve or publish. Unverified claims, stale versions, rejected reports, missing approval, tampering or incomplete writes stop export.	<i>reporting.py</i> ; approval and export tests. IN PLACE Offline; human result unavailable.
Prototype exclusions Charter P0/P1/P2	Engineering evaluation of controlled intake, evidence, verification and local review for RQ1 and RQ2.	Fixed rules for each stage, clear failures, source records and local audit make unsupported actions visible instead of quietly expanding scope.	Live connectors, participants, restricted-data study or deployment may proceed only through named future gates; they are not completion criteria for this slice.	No supplied credential use, production tenancy, social scraping, unrestricted crawling, autonomous publication, investment decisions, fine-tuning, cloud deployment or native PPTX/PDF generation.	Project Charter “Explicit non-goals”; requirements P1/P2; architecture gaps. CONTRACT BOUNDARY ; not an empirical result.

Source note: Developed by Faye Niu from versioned repository requirements, accepted design decisions and local checks. NIST (2023, PDF pp. 32–35) supports documented scope, oversight and independent assessment; Greshake et al. (2023, PDF pp. 1, 3–5) supports treating retrieved content as an untrusted instruction boundary; and Amershi et al. (2019, PDF p. 3) supports clear capability limits and efficient human correction. These sources do not certify this system. The matrix is a design summary, not a study result. Built means named code plus local checks, not live-source completion, ethics approval, restricted-data authority, security effectiveness or production readiness.

Table C.2. Fixed source connectors and their current states

Connector / source	Approved function / version	Identity, cut-off and period rule	Source location and saved copy	Outcome states	Current status / limit
Connector ability check	Source list plus policy version; exact rules for identifiers, facts, events, file types and collection modes. Derived-data rule <i>v2</i> .	Reviewed source-scoped identifier must belong to the company and be valid at cutoff. Cumulative facts require programme start through cutoff.	A fingerprint links the source and version, identifier, cut-off, programme window, requested facts and mode. Source-copy and derived-data SHA-256 values are checked on replay.	Before a call, blocked or invalid is different from a saved record , no-record , unavailable or failure . A rule-version mismatch stops the call.	BUILT TESTED OFFLINE. A traceable and consistent run does not prove the source is true.
Companies House replay	Companies House source; source list <i>1.4.0</i> ; policy <i>2026-08-26.1</i> ; offline replay.	Exact reviewed Companies House number. Filing, charge, incorporation and dissolution events must occur by the cut-off. Account states remain separate.	Write-once test bytes, checksum and source version; exact JSON locations and stable filing and charge identifiers; versioned derived-data hash.	Record; exact-ID no-record; unavailable source copy; invalid ID, duplicate or rule set; final failure kept separately. A blank is never changed to zero.	BUILT FICTIONAL REPLAY. No live registry result, current legal truth, complete coverage, valuation or production claim.
UKRI replay	UKRI source; source list <i>1.3.0</i> ; policy <i>2026-08-26.1</i> ; offline replay.	Exact UKRI organisation ID. Latest version available by cutoff is used. Cumulative totals require the exact programme-start-cutoff window.	Write-once test checksum; stable project and outcome-ID locations; the derived-fact location records the period, included projects, method and schema version.	Record or exact-ID no-record; invalid scheme, period or rules; failure remains separate. Missing or non-GBP amounts create clear incomplete states, never a zero or complete total.	BUILT FICTIONAL REPLAY. No live UKRI access, complete coverage, cause-and-effect or source-truth claim.
Companies House API mode	Read-only API mode appears in source list <i>1.4.0</i> , but its live-use flag is false. Licence and terms evidence is on hold.	An exact reviewed number, company link and cut-off would still be required. An API key or valid number format would not give permission.	No approved live copy exists. A future run would need write-once response bytes, source details, exact locations, and reviewed evidence for the source and version.	Blocked before connector invocation. Invalid configuration or identity remains invalid; no live record, no-record, unavailable or failure evidence exists.	HELD LIVE-UNRUN. Code presence is not permission, official retrieval evidence, availability or production readiness.
Public-web discovery / capture	Separate public-web research v1: find possible web addresses and collect pages safely. It is not a registry connector or a replacement for official source approval.	Public case, reviewed Companies House number, fixed cut-off and agreed claim or measure scope. Downloaded text stays outside the instruction boundary.	Search output is only a list of possible sources. Evidence needs accessible captured bytes, checksum, publisher and web address, exact passage and a source-specific decision.	Possible, collected, blocked, unsupported or failed gaps remain clear. Missing, held or conflicting evidence cannot become support. No search result is not an official no-record finding.	CONTROLLED ROUTE NOT USED IN A LIVE STUDY. Stop controls and local checks do not prove coverage, source truth, quality or live availability.

State rule: “record” means a successful collection containing an accepted record; it is not a truth label. No-record, unavailable, blocked, invalid and failure retain different meanings and reporting consequences. *Source note:* Developed by Faye Niu from versioned repository rules and local fictional tests. Nikiforova et al. (2020, PDF pp. 1–3, 15–17) supports use-case-specific quality checks; Gao et al. (2023a, ALCE, PDF pp. 3–4, 10) separates traceability/citation support from answer correctness; NIST (2023, PDF pp. 33–35) supports documented test conditions, uncertainty and limits. These sources do not certify a connector, publisher record or live-source result. The matrix is a design summary, not a coverage, quality, availability or production result.

Table C.3. Failures, recovery controls and current status

Area	Failure kept visible	Recovery or control	Evidence available
Identity and intake	Invalid, ambiguous or conflicting identifier; duplicate contract disagreement.	Named resolution; exact compatible repeats reuse the existing case.	Implemented; controlled tests. No live registry truth.
Fixed sources	Policy block, no record, unavailable source, invalid rules and final failure remain separate.	Offline checksum replay; stop on an unsupported function, identity, cut-off or version change.	Built; fictional replay. Direct live sources held.
Public-web research	Search, collection, extraction or report writing may be blocked, cancelled, reach its limit or fail.	Limited retries and use, stop controls, fixed task fingerprint and recovery of interrupted work.	Built; controlled tests. Live check and fixed comparison not run.
Claims and conflicts	Unsupported, old, untrusted, insufficient or conflicting evidence is not accepted.	Exact passage and source check; separate status; named decision on conflicts.	Built; fictional and controlled tests. Real-company effectiveness not measured.
Review and export	Old version or hash, incomplete verification, changed content or partly written output blocks release.	Named decision, removal of approval after an edit, version lock, file-list and hash checks.	Built; controlled tests. No automatic or remote publication.
Evaluation and operation	C0 and C3 observations, live company results, production availability and disaster recovery are absent.	Keep blank and held states. Require separate permission and a fixed study plan before running them.	Planned or outside scope. No outcome claim.

Source note: Developed by Faye Niu from the repository requirements, design, evaluation plan, source register and controlled test evidence at the Section 5 snapshot. Status describes executed engineering evidence only; it is not a production, availability, source-truth, user-benefit or investment-performance result.

Dataset, Source, and Freeze Manifests

Table D.1. C0–C3 evaluation-condition matrix

Condition	Workflow	Independent verifier	Human involvement	Scoring point	Fair-comparison controls	Current evidence state
C0 Manual	An authorised practitioner completes the current workflow in the fixed information window. Every tool or model used is logged.	No condition-level verifier.	Practitioner produces the report; there is no later C3 review stage.	Completed manual report; timing and corrections only if authorised.	Same measure definitions, period, eligible input, source-access window and agreed reference answers.	PLANNED / ON HOLD Waiting for permission, reference answers and a fixed study plan.
C1 Fixed single-agent	One extraction and report-writing route uses the shared catalogue and cleaning rules. Source records are kept.	No.	No human edits before scoring.	Automated C1 report before any review.	Shared controls at left, plus fixed schemas, prompts, code, environment and stop rules.	IMPLEMENTED Synthetic labelled D0 cases; mechanism evidence only.
C2 Controlled multi-agent	Separate roles collect, extract and clean data. The verifier checks support before fixed report writing.	Yes: separate from extraction and composition.	No human edits before scoring.	Verifier-gated C2 report before human C1 review.	Same shared and fixed technical controls as C1.	IMPLEMENTED Synthetic labelled D0 cases; mechanism evidence only.
C3 C2 + HITL	C2 output enters the versioned named-review workflow.	Yes: inherited from C2.	Named reviewer may edit, approve or reject under the authorised protocol.	Approved post-review version, scored separately from pre-review C2.	Same shared controls and C2 set-up, plus the fixed review plan.	WORKFLOW BUILT Human results are planned and on hold.

Fair-comparison rule: conditions use the same measure definitions, reporting period, eligible input, source-access window and agreed reference answers. Any tool or model used in C0 must be recorded. *Exclusion:* the public-web company-research route has not run live. It is not a C0-C3 condition and provides engineering evidence only. *Status:* developed by the author from the fixed project rules, with no study results or pass thresholds.

Table D.2. Agreed reference answers and equal source access

Stage	Required action	Required record or output	Reason to hold the study
1. Define Codebook + source policy	Create a versioned labelling guide from the data dictionary, with examples for measures, claims, sources and periods. Define which sources may be used and the cut-off date.	Labelling-guide version and hash; label schema; source rules and source-list template.	Hold labelling and scoring until the scope, examples, source rules and cut-off are clear.
2. Label Work separately	Two authorised reviewers with relevant subject knowledge label the same fixed set of sources independently. Keep their records separate until agreement is measured.	Reviewer IDs; source-set version; separate structured labels and times.	Unauthorised reviewer, different source set or discussion before the separate labels are complete: HOLD .
3. Record Complete label	Record the agreed value or reason it is missing, whether evidence is eligible, whether a source can be found, the period, claim support and required report section.	Structured label with source and location, plus a clear missing, unclear or conflict state.	Good writing or a citation does not prove support. Unsupported or unclear items remain held.
4. Measure Before discussion	Report raw agreement. Add a statistic corrected for chance only where the sample, label scale and task design support its use.	Total number of eligible labels; direct agreement; suitable adjusted measure, assumptions and uncertainty; otherwise NULL .	Do not import a pass threshold. High direct agreement can hide an uneven label balance, bias or differences in how text was divided.
5. Agree final labels Give reasons	Compare each disagreement with the fixed passage. Record the reason for the final choice and any change to the labelling guide.	Final label; reason; type of disagreement; revised guide where needed.	No silent change, no assumption that a majority is true, and no change to a definition without a new version.
6. Fix Reference + source list	Fix the final reference answers, labelling guide, source set and source-list hash before scoring any condition.	Write-once reference ID, version and hash, plus the scorer's input list.	Manual or system output is never assumed to be true. A later change needs a new version and a record of the change.
7. Score Blind where practicable	Score conditions against the same fixed reference without showing the scorer the condition where practical. Record any case where the condition becomes known.	Scores for each condition linked to the reference; record of blind scoring and any departure from it.	C0 manual and C1-C3 system outputs are conditions being tested, not the reference answer.
8. Keep equal Sources + time	Give conditions the same definitions, period, eligible source window and reference answers. Record inaccessible, deleted, paywalled, sign-in-only and later sources. Keep publication and availability separate from collection time.	Company-period source list; source state; publication, availability, collection and cut-off times; any difference in access.	Unequal access or support after the cut-off: hold the claim and report the difference. Never change missing evidence to zero.

CURRENT STATUS: REFERENCE ANSWERS HAVE NOT BEEN CREATED OR FIXED; THERE IS NO STUDY SCORE. *Source note:* Developed by the author from the current evaluation plan and data dictionary, with locally stored Artstein & Poesio (2008, PDF pp. 2–8, 16–23, 36–37), Gao et al. (2023a, ALCE, pp. 3–5, 8–10), Pineau et al. (2021, pp. 1–5, 9–13), Gebru et al. (2021, pp. 2, 4–10) and NIST (2023, pp. 32–35). This is a plan, not a result. It does not supply a result value, sample size or pass threshold.

Source Admission and Provider Data Controls

Table E.1. Ethics, leakage and repeatability controls

Risk	Control	Record kept	When to stop or ask for help	Current status
Restricted, personal or secret data is sent without permission	Classify data at intake and again before an external provider. Keep restricted and internal material local and out of external processing. Use private file permissions and record IDs, counts and hashes rather than raw values.	Data class, purpose and person responsible; source-copy and export checksum and file list; review decision; record without content; security check.	Unclear authority, unexpected personal data, an exposed credential or an unapproved processor: stop collection, isolate and record the event, then seek supervisor, ethics or security direction.	CONTROL BUILT / TESTED. Permission for real data, retention and deletion times, and confirmation that exposed keys were changed are NOT VERIFIED . No restricted external run is claimed.
Unauthorised C0 or C3 observation or reviewer feedback	Require recorded ethics and data permission, consent and withdrawal rules, approved participant information and questionnaire, coded IDs, minimum data and a quotation plan set in advance.	Approval or waiver; dated information and consent versions; separate ID link; task, review and withdrawal records; hashes before and after review.	If approval, consent or withdrawal handling is missing or unclear, do not recruit, observe or collect. Keep the result blank and ask for guidance.	APPROVAL NOT VERIFIED / ON HOLD. Human C0 and C3 work and feedback are unavailable. No participant result is claimed.
D0, D1 or D2 information leaks across study stages	Accept D0 by hash and test name. Use D1 only after permission. Keep D2 sealed with no visible path until the evaluation choices are fixed. Keep company and period groups separate. A change based on D2 makes the run exploratory.	Dataset list, hash and source version; group and period split; code, rule, prompt, model and analysis versions; loader result; access log and changes from the plan.	If the seal, checksum or group split fails, or D2 is viewed or used for tuning, stop the final study. Record the change and ask the supervisor to review it. Any later run is exploratory.	D0 CONTROL BUILT / TESTED. D1 has not run. D2 is sealed and blocked by the loader. No D1 or D2 result is claimed.
Public source is used unlawfully, is unavailable or is too late	Use versioned source rules, reviewed identity and cut-off, GET-only access with clear limits, robots and access checks. Do not bypass forms, sign-in, paywalls or CAPTCHAs. Review licence and sharing rules for each source.	Source ability and terms with review date; request fingerprint; availability, collection and cut-off times; copy hash and exact location; inaccessible or held state.	Unknown or changed terms, forbidden access, after-cutoff support or redistribution uncertainty: hold the source or claim; do not fetch or export; require named review.	LIMITED CONTROLS BUILT. Local public-web use is conditional. Direct live registry evaluation and sharing remain on hold or unavailable.

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Table E.1 continued: ethics, leakage and repeatability controls

Risk	Control	Record kept	When to stop or ask for help	Current status
Prompt injection or unauthorised model use	Treat downloaded text as data and reject content that looks like instructions. Allow only chosen public or synthetic input, set “store=False”, require a strict schema and exact passage, limit models and attempts, and never give the model approval or export rights.	Data class; provider, model and schema versions; attempt, token and error logs; exact passage and verification; reason for rejection or no claim; file list without content.	Injection, personal or restricted content, arbitrary routing, unclear processor authority or failed span validation: reject or abstain before claim support; stop model use and export.	CONTROL BUILT / TESTED. D1, D2 and participant processing have not run. There is no result for model quality, provider retention or better performance.
Version or setting changes; incomplete record	Fix the code revision and changes, dependencies, database version, source and dataset lists, cut-off, catalogue, labelling guide, prompts, schemas, connectors and models. Remove secrets. Keep run and agent IDs and hashes for analysis and exhibits.	Repeatability record; write-once input and result hashes; environment and database version; attempts, failures and human decisions; changes from the plan; hashes for generated files.	A missing or mismatched hash or version, secret in the set, unexplained change or failed repeat stops the comparison. Keep the failure and resolve it before making a claim.	MECHANISMS BUILT / TESTED. D0 repeat evidence exists. There is no final study record because the held conditions have not run.
Reference bias; builder and evaluator bias; selective reporting	Plan a versioned labelling guide, two authorised people working independently and a recorded final decision. Keep scoring blind where practical. Set outcomes and calculation totals in advance. Keep exclusions, failures, blank and negative results, and limit conclusions.	Separate labels; agreement, final decisions and labelling-guide versions; reference and scorer hash; case flow; analysis, changes from the plan and limits.	No authorised fixed reference, a rule changed after seeing outcomes, an unavailable calculation total or an unresolved disagreement means no final score. Mark the run exploratory or held and seek supervisor and statistical review.	DOCUMENTED PLAN. Reference answers and the final study plan are not fixed. C0, C3, D1 and D2 have no study results.

Status key: **built/tested** means repository code plus selected local checks run the stated mechanism; **documented plan** does not mean it was run; **approval not verified** means no claim of approval or refusal is made; **unavailable/sealed** cannot supply a result. *Source note:* Developed by Faye Niu from the current draft evaluation plan, governance rules and repository contracts, using local NIST (2023, PDF pp. 32–35), Gebru et al. (2021, pp. 2, 4–10), Greshake et al. (2023, pp. 1, 3–5), Kapoor & Narayanan (2023, pp. 1–7, 17–21) and Pineau et al. (2021, pp. 1–5, 9–13). This is a plan, not a result. The controls reduce risk but do not prove ethics compliance, lawful source use, security, repeatability, correctness or effectiveness.

Supplementary Evaluation Results

Table F.1. Measures, calculation totals and interpretation rules

Measure / group	Unit	Value counted	Total or comparison	Null / zero rule	Grouping and meaning
Claim precision	Scored emitted claim	Supported emitted claims.	All emitted claims eligible for scoring.	No eligible emitted claims: NULL / N/A with reason, never zero.	Treat claims from the same company or report as related; interpret with recall.
Claim recall	Required reference claim	Required reference claims produced correctly.	All reference claims required by the fixed scope.	No required labelled reference claims: NULL / N/A , never zero.	Group by company and report; missing claims remain visible.
F1	Condition + fixed claim set	$2PR/(P + R)$, using the precision and recall above.	Same fixed scoring set; no separate total.	NULL unless P and R are defined and $P + R > 0$.	Report P and R separately; F1 is not an acceptance threshold.
Unsupported-claim rate	Emitted claim	Produced claims without reference or evidence support.	All emitted claims.	No emitted claims: NULL / N/A , never zero.	Fix the support rules before scoring. Good writing or a citation does not prove support.
Claim-support rate	Emitted claim	Produced claims that meet the fixed support rule.	All emitted claims.	No emitted claims: NULL / N/A , never zero.	Keep separate from answer correctness and complete source records.
Contradiction recall	Reference conflict	Reference conflicts flagged correctly.	All reference conflicts.	No labelled reference conflicts: NULL / N/A , never zero.	Group linked conflicts; report false conflict flags separately.
Identity, acquisition, temporal + source quality	Company–source–period layer	Correct identity, acquisition state, temporal eligibility and source-quality outcomes, each separate.	TO DECIDE BEFORE STUDY: units with labels for each layer.	A layer with no labels is NULL ; event outcomes are NULL in current D0.	No single quality score; group by company, report and whether a source can be found.
Schema + extraction	Extraction record / field	Output that follows the schema; exact or field-level match to the fixed label.	TO DECIDE BEFORE STUDY: attempted and eligible records, and record-level or field-level scope.	No labelled eligible records: NULL , never zero.	Report by format, measure type and whether a source can be found. Fields in one report are related.
Normalisation + missing state	Company–metric–period field	Correct type, value, unit or currency and correct missing-state label; retain blank–zero matrix.	TO DECIDE BEFORE STUDY: labelled fields in scope by task and data type.	No labels: NULL . Zero is a valid observation, not missing; zero accuracy requires a positive denominator.	Report by data type/missing class; do not pool repeated company fields as independent.
Complete source records	Eligible emitted claim	Claims with source ID, locator, applicable publisher, retrieval time, checksum and period.	All eligible emitted claims.	No eligible emitted claims: NULL / N/A , never zero.	Completeness makes a claim traceable. It does not prove that the source or claim is true.

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Table F.1 continued: measures, totals and interpretation rules

Measure / group	Unit	Value counted	Total or comparison	Null / zero rule	Grouping and meaning
Source-location validity	Claim and location in a fixed source copy	Exact location opens the accepted fixed source copy.	TO DECIDE BEFORE STUDY: eligible locations and claims, and how several locations are handled.	No eligible inspected locators: NULL ; a broken inspected locator is a failure, not missing data.	Keep resolution separate from passage support and source truth.
Report-section completeness	Report + required section	Required section status against the fixed reference template.	TO DECIDE BEFORE STUDY: required sections and how held or empty sections are handled.	No fixed template or scored report: NULL .	Aggregate at report/company level; presence does not establish factual correctness.
Factual errors + severity	Scored claim / report	Error count using categories set in advance; seriousness label for each error.	Count per report; the rule for combining seriousness is TO DECIDE BEFORE STUDY .	No scored report: NULL . Zero only after an eligible report is checked and no error is found.	Fix seriousness definitions before the scorer sees the condition or outcomes.
Transparency rate	Missing, conflicted or stale item	Eligible items surfaced with their state and reason.	TO DECIDE BEFORE STUDY: exact eligible items and how report-level results are combined.	No eligible labelled items: NULL / N/A , never zero.	Report missing, conflict and old-information groups separately. Do not use one score without context.
Human time	Authorised human + report	Active human minutes and elapsed wall-clock minutes, recorded separately with common start/stop rules.	Match the same company, period and report under one “review-ready” definition. How results are combined is TO DECIDE BEFORE STUDY .	Current D0 and unauthorised C0/C3: NULL ; never reconstruct missing time as zero.	Group repeated tasks by reviewer and report. Record tools, interruptions and waiting.
Machine stages + retries	Stage, attempt and run	Stage duration, attempt/retry and failure/re-run events.	Executed stage or run; summary method and a comparison that includes failures are TO DECIDE BEFORE STUDY .	Unexecuted stage: N/A with reason, not zero.	Pair same company-period; retain failures and stage boundaries.
Tokens + monetary cost	Model, attempt and run	Input/output tokens by model/run; cost from official execution-date price, recorded formula and currency.	Same executed workflow unit and model/version; do not estimate an absent price.	Missing logs or price: NULL . Zero only when logs prove there was no billable use.	Report tokens separately from cost; version and date bind the comparison.
Repeat reliability	Fixed configuration + repeated unit	Complete equality for fixed outputs; field or claim agreement for settings that can vary; completion, schema validity and failure categories.	TO DECIDE BEFORE STUDY: repeat set, output scope, matched total and agreement measure for settings that can vary.	No qualifying repeats or denominator zero: NULL , never zero.	Group by company and report. Show failures and configuration hash. There is no universal pass level.
C3 usability + edit burden	Authorised reviewer + task/report	Ethics-approved task success, time, changed decisions, whether confidence matches correctness, or a licensed scale; edit events and optional text or meaning-based distance.	TO DECIDE BEFORE STUDY in the authorised questionnaire and task plan. Keep C2 before review and C3 after review.	Current D0 with no participant observation: NULL . Do not invent reviewer value.	Group by reviewer and report. Record order and experience. Do not choose a scale after seeing outcomes.

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Table F.1 continued: measures, totals and interpretation rules

Measure / group	Unit	Value counted	Total or comparison	Null / zero rule	Grouping and meaning
CURRENT STATUS: PLANNED. THERE IS NO STUDY RESULT OR PASS THRESHOLD.					
Universal rule: no eligible labelled cases or a zero denominator yields NULL / N/A with a reason , never a measured zero. <i>Source note:</i> Developed by the author from the current evaluation plan and data dictionary, using local Gao et al. (2023a, ALCE, PDF pp. 3–4, 10), Nikiforova et al. (2020, pp. 1–3, 15–17), Pineau et al. (2021, pp. 1–5, 9–13), Artstein & Poesio (2008, pp. 2–8, 16–23, 36–37), NIST (2023, pp. 32–35) and Amershi et al. (2019, pp. 3–5, 16–19). These sources motivate separate and documented measures that include human work. They do not validate this system or supply its thresholds.					

Table F.2. Complete D0 fictional-case decision ledger

Case	Category	Candidate	Evidence state	Expected decision	C2 consequence
complete_internal_support	Complete	Employees = 12	Matching submission	Supported	Emit
sparse_not_reported	Sparse	Not reported	No evidence required for a claim	No claim	Hold
zero_is_observed	Blank / zero	Jobs = 0	Matching submission zero	Supported	Emit
blank_is_missing	Blank / zero	Empty value	No evidence	No claim	Hold
mixed_type_integer	Mixed types	Employees = "17"	Matching integer	Supported	Emit
conflicting_public_value	Conflict	Grant = GBP 100	Same-period evidence = GBP 200	Contradicted	Hold conflict
stale_public_value	Stale	Awards = 1	Matching value in prior period	Stale	Hold
unsupported_no_evidence	Unsupported	News items = 4	No evidence	Insufficient evidence	Hold
ambiguous_company	Identity	Employees = 8	Matching submission, identity unresolved	Identity hold	Hold
duplicate_same_value	Duplicate	Grant = GBP 300	Two matching sources	Supported	Emit
inaccessible_source	Inaccessible	Grant = GBP 700	Source unavailable	Insufficient evidence	Hold
narrative_structured_fact	Narrative	Products = 2	Matching narrative evidence	Supported	Emit
mixed_currency_conflict	Mixed currency	Investment = GBP 400	Evidence = USD 400	Contradicted	Hold conflict
prompt_injection_evidence	Prompt injection	Partnerships = 5	Matching value marked untrusted	Rejected untrusted	Hold

Source note: Exact case identifiers, categories, expected states and values from `fixtures/evaluation_cases.json`. The ledger is synthetic and does not describe any company.

Table F.3. Controlled company-research and hostile-input results

Area checked	Cases exercised	Result	Limit
Full research route	Reviewed identity, limited discovery, capture, exact passage, v3 report, named review and download	PASS	Fake or local evidence; no result for live accuracy, coverage or usefulness.
Task and budget rules	Fixed fingerprints, call budget including retries, time limit, zero redirects, cancellation and recovery after interruption	PASS	No live result for delay, tokens, provider limits or cost.
Network protection	HTTPS, credential and port checks; DNS changes; redirects; robots rules; file type and size limits; connection pinning and TLS	PASS	Selected threats only; not a complete server-side request forgery or production security test.
Evidence checks	URL matches the source list, run ownership, duplicate removal, exact passage and supported-value checks	PASS	A supporting passage does not prove publisher truth or complete discovery.
Source mix and conflict	Sources rotated by publisher, balanced batches, and preserved contradictions and gaps	PASS	No estimate for real publisher coverage or recall.
Review and cleanup	Old or changed decisions rejected, source ownership checked, and one run cannot clear another	PASS	No result for reviewer benefit or several users working at once.
Focused suite total	37 tests; one dependency deprecation warning	37 PASS	Controlled local execution only; all external calls absent.

Source note: Fresh local execution of company-research integration, database update and fictional-case tests. PASS means that the named check passed. It does not approve a source, model, company profile, human process, network boundary or production deployment.

Table F.4. Planned public-company source-discovery comparison

Comparison element	Required fixed evidence	Current state	What can be reported
Units and identity	Authorised public-company list; reviewed Companies House number; company cut-off date	HELD	No sample size, company result or wider claim.
Equal source access	Eligible-source list, availability states and the same cut-off window	HELD	No fair comparison between official-only and assisted routes.
Official-only condition	Approved live official connectors and write-once captures	HELD	Fictional replay cannot replace live official evidence.
Assisted condition	Authorised count, checksum and locator check; fixed query, model, prompt and budget; protected capture	BUILT; NOT RUN LIVE	No live discovery, delay, token or cost value.
Independent reference	Relevant claims, exact passages, conflicts, gaps and unavailable sources, fixed before scoring	NOT CREATED	Precision, recall and category coverage remain unavailable.
Planned measures	Support rates, source records, conflicts, source and category coverage, gaps, time, retries, tokens and cost	PLANNED	No value is reported because no suitable total exists.
Interpretation	Paired comparison for the same company and cut-off, with stated budgets, missing data and failures	FUTURE	It could show limited added discovery, but not a complete web search or investment quality.

Source note: Author summary of the fixed study plan, approved-source list and current implementation. This table describes future work. It contains no result for a company, model, source search, cost, time or usefulness.

Table F.5. Manual and human-in-the-loop comparison status

Condition	Built or planned	Missing permission or evidence	Current result
C0 manual	Planned same-source period, clear stopping point, active and total timing, and a plan to record tools and corrections	Authorised practitioners, ethics and data permission, fixed tasks, independent reference answers and sample basis	0 cases; all quality, time, cost and usefulness measures NULL .
C2 before review	Fixed verified output can be saved and scored before a person sees it	No authorised real-company or reference dataset	D0 mechanism results only; no human comparison result.
C3 HITL	Versioned interface, edit history, approval, rejection and reasons	Authorised reviewers, consent and study form, tasks hidden from the reference answers, order plan and observed events	0 cases; edits, reversals, timing, quality and reviewer usefulness NULL .
Reference process	Independent labels from two reviewers, agreement, a way to settle differences and a fixed version are planned	No completed labels, observed disagreements or final decisions	Agreement and reference-quality measures NULL .
Future-pilot consequence	The review design and planned paired measures can be described	No human observations	No observed claim about time saving, error correction, usefulness or decision quality.

Source note: Author summary of the fixed study plan and current evaluator results. The table reports that the human study was not run. It does not show a negative effect, and no participant data were collected or inferred.

Table F.6. Research-question evidence and answer status

Question	Available evidence	Answer supported by the evidence	Missing evidence or limit
Primary question	Implemented local workflow; engineering tests; D0 C1/C2 comparison	ANSWERED WITHIN CONTROLLED SCOPE: evidence records and the verifier effect are demonstrated	No inference to real-company accuracy, business benefit or production use.
RQ1	Identity, time, source, verification and export controls; fourteen designed D0 cases; three repeats	IMPLEMENTATION AND CONTROLLED BEHAVIOUR: the required states and links are retained	No estimate for real-world accuracy or source coverage.
RQ2	Paired C1/C2 D0 output; five supported and six unsupported emitted candidates; zero local LLM cost	OBSERVED D0 EFFECT: the verifier held the six designed unsupported candidates without reducing D0 recall	Visible labels helped shape the test, false rejection outside D0 is unknown and timing is not useful.
Future pilot	Manual baseline and versioned named-review controls are defined	NOT A CURRENT RESULT: the design is prospective	No participant observations, active time, cost, editing or usefulness evidence.

Source note: Author summary of Chapters 1, 3 and 5. The status labels separate what was built, what was measured with fictional cases and what is proposed for the pilot.

Reproducibility and Build Manifest

Table G.1. Implementation and evidence record at 28 August 2026

Record item	Recorded value	Meaning
Host/runtime	Darwin 25.6.0 arm64; Python 3.12.13	Local execution environment; not a portability result.
Git state	HEAD 6356aa49cb3b; dirty working tree recorded	The commit is a base reference, not a complete content seal.
Python dependencies	pyproject.toml SHA-256 d8ed0555...; runtime lock 5cce8f8e...; development lock 5e2469bc...	The declared and installed dependencies are recorded with checksums.
Database	Alembic revisions 0001–0009; current head 0009	Fresh SQLite upgrade and schema-drift check executed.
Evaluation list	SHA-256 1c16234a...; schema evaluation-manifest-v2	D0 was run; D1 is only planned; D2 remains sealed.
D0 cases	14 cases; fixture SHA-256 f403d59e...; namespace benchmark:d0	Fictional, labelled cases used only for engineering tests.
D0 output	3 repeats; output SHA-256 e3be8f62...	The current result shows how the mechanism behaves.
Evidence not collected	C0 and C3: zero observations; live company/model/web: not run	There is no evidence yet about people, live sources, usefulness or production use.

Source note: Local commands, repository checksums, database update output and D0 evaluator output recorded by the author. The shortened checksums are shown in full in the chapter and repository records. This information helps repeat the work, but it does not prove that the sources are true, that the findings apply elsewhere, or that the system benefits users.

Table G.2. Engineering check results recorded on 28 August 2026

Check	Result	Evidence from the run	Limit
Full automated suite	PASS	286 passed; 22 warnings; 85.58% statement coverage	Fictional, fake and local inputs; coverage does not prove that the software is complete or free from faults.
Ruff lint	PASS	No lint finding across src and tests	Only the configured static rules were checked.
Ruff format check	FAIL	Five files would be reformatted: two source and three test modules	Dated result only; later code changes were not evaluated by this record.
Strict MyPy	PASS	No issues in 46 source files	Only typed code and the configured rules were checked.
Database migration	PASS	Fresh SQLite upgrade 0001–0009; snapshot head 0009; no pending operations	Local SQLite only; production concurrency and recovery were not tested.
Diff whitespace check	PASS	git diff –check returned success	This does not check meaning or untracked files.
External/live actions	NOT RUN	No company, model, registry or public-web call	Live availability, coverage, latency and cost remain unknown.

Source note: Fresh local commands were run on the dated version recorded in EVAL-T1. PASS applies only to the named check. The formatting failure and warnings are reported because they are part of the result.

Supplementary Exhibits and Text Alternatives

Table H.1. Conditions for use in another setting

Area	What must be checked again	Current evidence	What cannot be concluded
Data meanings	Measures, periods, missing states, whether sources can be found, currencies and cut-off dates	Versioned local rules; fourteen D0 cases	Accuracy for new companies, documents, sectors or periods.
Identity and sources	Official identifiers, licences, availability, source levels and exact locations	Offline replay and controlled fake web sources	Current official truth, live coverage or stable access.
Model and search	Provider, model and prompt versions; ranking; budgets; retention; and failure behaviour	Protected route built but not run live	Future model or search quality, delay or cost.
Human setting	Intended reviewer, permission, task, interface accessibility, workload and effects on decisions	Named control mechanisms; zero C0/C3 observations	Time saving, error reduction, usefulness or decision quality.
Operation and security	Sign-in, separation between users, storage, network access, monitoring, backup, incidents and recovery	Local single-user controls and controlled hostile-input tests	Hosted, multi-user, reliable or production-secure operation.
Decision use	Permitted purpose, communication review and professional accountability	No recommendation, scoring or valuation functions	Investment merit, prediction, advice or replacement of professional judgement.

Source note: Author summary of the tested version and the remaining conditions. The table lists what must be checked before using the system elsewhere. It is not a deployment checklist, a compliance statement or proof that the move would work.

Table H.2. How the results compare with the literature (qualitative, not a non-anomaly test)

Question	Expectation from the literature	Observed evidence	Comparison
RQ1	Clear source records, data-quality rules and repeat controls make decisions easier to inspect, but do not prove that a source is true or that the system is accurate in real use.	Identity, time, source, verification and export controls ran; fourteen D0 cases repeated exactly.	Compatible in direction: this only shows that the controlled mechanism can run. It does not conflict with the reviewed literature, but there is no real-world accuracy estimate.
RQ2	Claim correctness, citation support and attribution should be assessed separately; a support gate should withhold represented unsupported material.	C2 removed six D0 false positives, retained all five supported cases and changed precision from 0.455 to 1.000 without D0 recall loss.	Compatible in how it works: the direction matches the design drawn from the literature, but the perfect values reflect the designed cases and cannot show a normal population effect.
Future human pilot	Human control may improve reflection and accountability, while also creating reliance, preference, workload and delay trade-offs.	Named review controls exist, but C0 and C3 each contain zero observations and every human outcome is null.	No observed comparison: the pilot design is prospective and no human effect is claimed.
Generalisation	Independent held-back evidence, protection against test leakage, complete records and clear uncertainty are needed before making wider claims.	D0 labels and rules were co-developed; D1 is protocol-only, D2 is sealed, and live/company comparisons are unrun.	Expected evidence gap: this agrees with the cautions in the methods literature, but it cannot support a statistical judgement that the result is anomalous or non-anomalous.

Source note: Author comparison of the Chapter 5 results with Gao et al. (2023a, 2023b), Pineau et al. (2021), Nikiforova et al. (2020), Gebru et al. (2021), Kapoor & Narayanan (2023), Amershi et al. (2019) and Bućinca et al. (2021). The sources motivate expectations and limits of the method. None provides a population distribution that can be compared directly with D0.

Table H.3. Governance controls, evidence and remaining requirements

Risk area	Current control	Current evidence	What is still required
Data and privacy	Classification, purpose declaration, local restricted/internal boundary, public-text redaction	Policies, schemas and controlled tests	Approved lawful-purpose, retention, deletion, backup and incident responsibilities.
External model	Public or fictional data only, reviewed route, strict schema, budgets and store=False	Code and fake-provider tests; no live request	Eligible account settings, observed request and retention evidence, and a current provider review.
Untrusted content	URL candidates only, protected collection, instruction filtering and exact-passage checks	Controlled malformed-content and grounding cases	Independent hostile-input testing and continuing maintenance of source rules.
Network transport	HTTPS, DNS, redirect, port, file type and size checks with public-IP pinning	Controlled SSRF, rebinding and redirect cases	Separate production network access, sign-in and separation between users.
Resource and recovery	Budgets, retries, cancellation, fixed fingerprints, checksums and staged export	Task and export recovery tests	Backup, service targets, monitoring, recovery after a forced stop, and incident exercises.
Decision and persons	No recommendation, valuation, suitability, profiling or reputation scoring	Requirements and output contracts	Separate authority for any changed purpose, audience or regulated decision use.
Research validity	Fixed rules, explicit missing results, source and dataset lists, and D0 repeats	Fourteen designed cases; D1/D2 and C0/C3 absent	Independent references, a sealed future evaluation and authorised human observations.
Operation and access	Local single-user deployment and meaningful printable output	Local tests and visual PDF inspection	Full accessibility audit, assistive-technology study and evaluation of a production design.

Source note: Author summary of the implemented controls, Chapter 5 evidence and declared holds. This table records the dissertation's evaluated state; it is not legal advice, a compliance assessment, a penetration test or deployment approval.

Table H.4. Ordered evidence plan after this dissertation

Stage	Activity requiring permission	Evidence needed to proceed	Result unavailable before this stage
1	Authorised matched business pilot	Fixed companies, periods, sources, reference decisions, completion rule, staff owners and acceptance criteria	Supported-claim quality, human time, edits, source failures, completion and cost per report.
2	Live check using only counts, checksums and locations; no report content	Named permission, approved routes, account and retention evidence, and a write-once run record	Evidence about live connection, retention route and basic operation.
3	Official connectors and a fixed public-company source benchmark	Fixed companies, cut-off date, sources, queries, budgets and independent passage reference	Estimates for added supported discovery, failures, resources and cost.
4	One sealed final evaluation	Unchanged study plan, no tuning on held-back cases and analysis that includes failures	Claim-quality and reliability estimates for new cases.
5	Managed-platform comparison	Equivalent inputs, source windows, output contract, scoring rules and recorded setup effort	Comparative setup, maintenance, traceability, failure recovery and total cost.
6	Accessibility, security and operational assessment	Assistive-technology audit, independent security scope, ownership, monitoring and recovery exercises	Any justified usability, resilience, security or production-readiness conclusion.

Source note: Author summary of missing evidence and stopping conditions. Each stage requires separate permission and a fixed study plan. The order does not imply approval, success or an obligation to deploy.

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